

ATLAS: Adaptive Trading with LLM AgentS Through Dynamic Prompt Optimization and Multi-Agent Coordination

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Abstract

Large language models (LLMs) offer promising capabilities for financial decision-making, yet their deployment in sequential trading settings faces two key challenges: synthesizing heterogeneous information sources and adapting agent behavior under delayed and noisy reward signals. We address these challenges by introducing *ATLAS (Adaptive Trading with LLM AgentS)*, a unified agentic framework for systematic integration of market data, financial news, and corporate fundamentals, and *Adaptive-OPRO*, a novel prompt optimization method that dynamically updates agent instructions using real-time stochastic feedback. We evaluate our approach across regime-specific equity trading scenarios and multiple LLM families. Results demonstrate that Adaptive-OPRO consistently outperforms existing methods, particularly in highly volatile regimes. Moreover, our analysis reveals that increased information availability does not necessarily translate to improved performance, highlighting the importance of careful modality integration in noisy market environments.

1 Introduction

Financial markets represent one of humanity’s most complex decision-making environments, requiring synthesis of vast information, from technical indicators and fundamental analysis to breaking news and market sentiment. LLMs introduce new possibilities for financial decision-making through their ability to process diverse data sources and reason over complex scenarios.

From the model’s perspective, financial trading serves as an ideal testbed: it combines unambiguous metrics, sequential complexity, multimodal reasoning requirements, and inherent stochasticity. Unlike synthetic benchmarks, markets provide extensive historical data without simulation bias and reward genuine understanding over pattern memorization. LLMs can therefore be tasked to make

decisions under uncertainty, revealing capabilities in complex reasoning (He et al., 2025), market understanding (Li et al., 2025), and high-risk decision-making (Hung et al., 2023).

Despite this potential, stock market decision-making introduces inherent challenges beyond stochasticity. Decisions require synthesizing heterogeneous signals such as price dynamics, market conditions, and firm-specific developments into coherent actions. Moreover, in high-stakes financial environments where capital is continuously at risk, static decision policies are insufficient; decision patterns must be revised by incorporating market feedback as it unfolds, enabling continuous behavioral adaptation.

Consequently, turning LLM capabilities into reliable trading systems raises two key queries: (i) how diverse signals are synthesized into coherent guidance, and (ii) how models adapt their behavior through continuous market interaction. While recent work explores these issues partly, their systematic study in realistic trading settings is limited.

In this work, (i) we propose *ATLAS*, a multi-agent framework that provides a foundational structure for experimentation in LLM-based stock market decision-making; (ii) we introduce *Adaptive-OPRO*, a prompt optimization mechanism for sequential settings that supports behavioral adaptation through ongoing market interaction and achieves state-of-the-art performance across multiple market regimes and LLM families. Through extensive regime-aware evaluations, we show that additional input modalities are not uniformly beneficial and depend critically on market conditions.

2 Related Work

LLM Agents in Financial Markets Recent work explores several LLM-based trading agents, from sentiment-driven (Kirtac and Germano, 2024) to coordinated, multi-component systems (Zhou

et al., 2025; Yang et al., 2025; Liu et al., 2023). Examples include CryptoTrade, which integrates on/off-chain signals with reflection (Li et al., 2024), and TradingAgents, which coordinates specialized roles via structured debate and synthesis (Xiao et al., 2025). Memory-centric designs such as Fin-Mem emphasize persistent, task-specific recall (Yu et al., 2023), while FINCON introduces conceptual verbal reinforcement to shape multi-agent collaboration (Yu et al., 2024). Others incorporate learning signals (Xiong et al., 2025) or mixture-of-experts routing (Ding et al., 2025), and focus on document-centric analysis, e.g., filings and earnings calls (Fattouros et al., 2025). However, key limitations persist: prompts are mainly hand-crafted even when feedback is delayed and noisy, and many setups collapse execution into directional scores. ATLAS pairs a prompt-tuning component with order-level evaluation (type, size, timing, price) in a simulator built for such interfaces (Papadakis et al., 2025), using multi-run reporting to account for stochastic variability (Song et al., 2025; Atil et al., 2025).

Prompt Optimization enhances LLM performance beyond manual tuning. Optimization by PROMPTING (OPRO) treats the model as a meta-optimizer over instruction text, achieving gains on single-turn tasks with immediate feedback (Yang et al., 2024). Extensions explore evolutionary search and reinforcement-style updates (Guo et al., 2025; Do et al., 2024; Austin and Chartock, 2024). These settings typically assume fast, unambiguous scoring and independent instances. In contrast, trading provides deferred, noisy reward signals and sequentially coupled decisions. Adaptive-OPRO addresses such challenges by using rolling evaluation windows and by separating static instructions from dynamic run-time content, allowing stability where consistency matters and controlled evolution where change is beneficial.

3 ATLAS Framework

ATLAS comprises three main components: (i) a *Market Intelligence Pipeline*, which consists of specialized agents that prepare market, news, and fundamental inputs for downstream decisions; (ii) a *Decision & Execution Layer* centered on a Central Trading Agent that generates and executes orders; and (iii) a feedback mechanism that collects post-execution signals and feeds them back for continuous adaptation. Within the feedback mechanism we incorporate Adaptive-OPRO, an extension of

the OPRO framework that dynamically edits the Central Trading Agent’s instruction prompt based on real-time, stochastic market feedback. Figure 1 provides an overview of the ATLAS framework.

Market Intelligence Pipeline. ATLAS separates information preparation from decision-making. The Market Intelligence Pipeline consists of three specialized agents, each with a distinct analyst role. **Market Analyst** produces multi-timescale summaries from price and volume in varying time scales (2 years, 6 months, and 3 months of history with monthly, weekly, and daily candlesticks, respectively). Within each window it computes standard indicators (e.g., moving averages, momentum, volatility bands, support/resistance) and refreshes daily, providing a consistent, noise-filtered description rather than trading signals (details in App. B). **News Analyst** aggregates relevant articles into structured fields (*Sentiment Assessment, Key Developments, Market Relevance, Source Analysis*) with optional full-text retrieval to move beyond headlines (details in App. C.1). **Fundamental Analyst** extracts material changes from periodic reports and corporate events, activating infrequently to mirror reporting cycles and provide medium- to long-horizon context (details in App. C.2).

Decision & Execution Layer. Here, the trading actions (e.g. buying or selling a stock) are determined, executed, and corresponding market feedback is received. The main decision-making component within this layer is the **Central Trading Agent (CTA)**. This agent consumes the structured inputs and current portfolio and emits orders that specify type (market, limit, stop), size, timing, and price levels. Orders are executed in StockSim (Papadakis et al., 2025), which enforces core trading semantics and returns fills, positions, and cash for the next step. Order-level decisions clarify intent and link analytical quality to execution choices.

Feedback Mechanism. This mechanism defines how information derived from market outcomes is incorporated into the agent’s future decisions. It may be entirely absent, resulting in a static agent that follows a fixed policy, or it may be enabled to support adaptation based on observed performance. In general, the mechanism processes signals such as returns or behavioral outcomes from past decisions and uses them to influence subsequent actions. Implementations can range from simple feedback summaries to more structured optimization

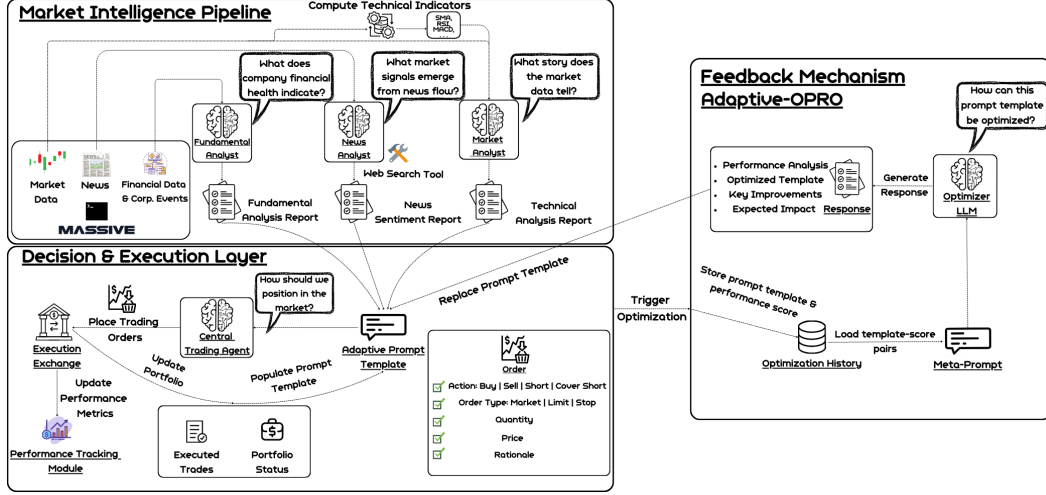


Figure 1: ATLAS Framework Overview. The Central Trading Agent submits orders to the Trading Execution Engine via prompts shaped by three specialized analysts and the proposed *Adaptive-OPRO* optimization technique.

approaches, such as reflection-based methods (Li et al., 2024). In the following section, we describe Adaptive-OPRO, a prompt optimization technique that leverages market feedback to iteratively refine the agent’s decision-making process.

4 Adaptive-OPRO

Adaptive-OPRO is a sequential prompt-optimization procedure that improves an agent’s instruction prompt using delayed, noisy performance feedback. It generalizes OPRO to interactive settings where decisions are temporally coupled and rewards arrive after multiple steps. The core idea is to treat instruction text as the optimized object and to update it periodically via a learned update mechanism (implemented by an optimizer LLM), while keeping the agent’s run-time inputs and interfaces stable.

Optimized object, state, and round inputs.

Adaptive-OPRO maintains a current instruction prompt P_t for a target agent that acts over time, along with an *optimization history* $\mathcal{H} = \{(P_i, s_i)\}_{i < t}$ storing past prompt variants and their scores. At the end of each evaluation window, it constructs an optimizer query with the following inputs: (i) a *meta-prompt* M that specifies the optimizer’s role and constraints, (ii) $\mathcal{H}$ (or a compact summary thereof), and (iii) a summary of the agent’s recent interaction outcomes together with a scalar performance score s_t for P_t . An update rule U , implemented using an optimizer LLM, produces a revised prompt $P_{t+1} = U(M, \mathcal{H}, s_t, \text{summary})$. The output of each round is an updated instruction prompt that governs subsequent agent decisions.

Stability via template separation. In sequential systems, prompt updates can inadvertently break the run-time interface (e.g., input placeholders, output schemas) or overfit to transient observations. Adaptive-OPRO therefore separates the target agent prompt into: (a) *static instructions* (policy, priorities, constraints, formatting requirements), and (b) *dynamic run-time content* injected at execution time (state, observations, tool outputs, recent actions). Only the *static instruction block* is editable; all placeholders and the run-time injection format are held fixed. This enforces *edit locality*: updates can change *how* the agent reasons and decides, but cannot change *what* information it receives nor the interface it must comply with.

Windowed evaluation under delayed feedback.

To address credit assignment and reduce variance, Adaptive-OPRO evaluates prompts over rolling windows of K decision steps. After each window, the system computes a scalar performance score s from outcomes observed during that window (e.g. task success, reward, utility, risk-adjusted return). The choice of K and scoring function is task-dependent; the only requirement is that s provides consistent ordering to compare prompt variants.

Meta-prompted update rule.

At each window, Adaptive-OPRO forms the optimizer query from M , $\mathcal{H}$, and the recent outcome summary/score, and applies U to generate a candidate P_{t+1} (Figure 2). The optimizer is instructed to: (i) diagnose likely failure modes of the current prompt, (ii) propose a revised instruction prompt P_{t+1} , (iii) summarize the concrete changes made, and (iv) state the

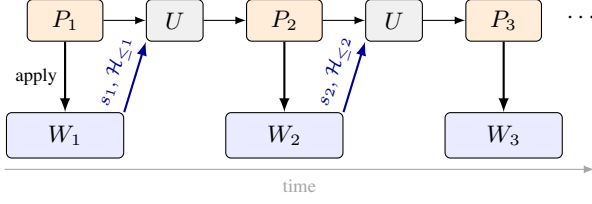


Figure 2: Adaptive-OPRO as an online procedure. Prompt P_t governs window W_t ; its realized score s_t and the accumulated history $\mathcal{H}_{\leq t}$ feed the optimizer U , which emits P_{t+1} for the next window.

expected behavioral impact. The candidate is accepted only if it preserves the template (e.g., placeholders and output schema). The accepted prompt governs the next window onward and is appended to history with its subsequent score; since no window is ever replayed under a later prompt, each variant is assessed on outcomes that unfolded after it was written, keeping optimization and evaluation on disjoint segments of the trajectory.

ATLAS instantiation. In ATLAS, Adaptive-OPRO is applied to the *Central Trading Agent*’s instruction prompt (i.e., the static instruction block of the decision policy). Dynamic run-time content corresponds to the daily injected analyst summaries, portfolio state, and recent executions, which are kept fixed by construction (App. I). For scoring, we aggregate portfolio performance over $K=5$ trading days to reduce noise and capture delayed effects of sequential decisions, then map cumulative ROI to a bounded OPRO-style score $s \in [0, 100]$ via linear scaling and clipping:

$$s = \text{clip}_{[0,100]}(50 + 250 \cdot \text{ROI}), \quad (1)$$

so that $-20\% \mapsto 0$, $0\% \mapsto 50$, $+20\% \mapsto 100$. This yields a stable, delay-aware signal while limiting the impact of outlier windows; the optimizer is restricted to instruction edits that preserve ATLAS’s execution interface.

5 Experiments

Our study examines ATLAS along three axes: **(1) Adaptation** – whether sequential prompt optimization via *Adaptive-OPRO* improves over well-tuned static prompts and over analytical reflection when feedback is delayed and noisy; **(2) Component attribution** – contribution of structured inputs (Market Analyst, News Analyst, Fundamental Analyst) under different regimes; **(3) Model capabilities** – performance of backbone LLMs as both decision

policies and prompt optimizers under Adaptive-OPRO, assessed by return and risk-adjusted performance, robustness across runs, and their ability to propose instruction updates for sustained improvements over windows.

5.1 Experimental Setup

Assets and time period. Specifically, we evaluate stock market decision-making across three distinct regimes: a **bearish-volatile** regime characterized by declining prices and elevated uncertainty, a **sideways** regime marked by range-bound price dynamics and limited directional trends, and a **bullish** regime defined by sustained upward momentum and comparatively favorable risk-return conditions. Each window spans two months (Apr 28-Jun 28, 2025) *with a daily decision interval*: the agent acts once per trading day. This horizon is chosen to (i) capture multiple decision cycles *without regime mixing*, so adaptation reflects outcomes rather than macro shifts, and (ii) preserve complete conversation history (analyst summaries, orders, prompt-evolution logs) within the context limits of all backbones, enabling fair, auditable runs across models and ablations. More details in App. D.1.

The experimental setup, including the evaluation method, metrics, regime partitioning, and evaluation horizon, follows Li et al. (2024), ensuring methodological consistency and fair comparison across settings. We explicitly account for LLM stochasticity by running each configuration *three times* and reporting mean $\pm$ standard deviation, distinguishing systematic performance differences from randomness rather than single-run variability.

Models. We evaluate seven backbones spanning families, sizes, and reasoning modes: GPT-o3, GPT-o4-mini, Claude Sonnet 4 with and without thinking, Llama 3.3-70B, Qwen3-235B, and Qwen3-32B. Each run uses a single backbone for all ATLAS components and Adaptive-OPRO, isolating how model capacity and architecture affect sequential behavior, instruction adherence, stability, and cross-family transfer without per-model tuning.

Prompting strategies. We compare three strategies for the *Central Trading Agent*: **Baseline** – a fixed instruction prompt obtained via iterative expert prompt engineering; **Reflection** (Li et al., 2024) – a weekly reflection mechanism that summarizes recent trajectories into high-level feedback that the agent must interpret; **Adaptive-OPRO** – our sequential prompt optimization with windowed scoring and template separation (Section 4). Our

Model	Prompting	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
Non-LLM-Based Strategies						
Buy & Hold	N/A	-8.59	-0.071	20.45	0.00	1
MACD	N/A	6.50	0.131	6.86	0.00	1
SMA	N/A	6.91	0.177	3.56	50.00	4
SLMA	N/A	-1.87	-0.078	6.89	0.00	1
Bollinger Bands	N/A	0.00	0.000	0.00	0.00	0
LLM-Based Strategies - ATLAS						
Llama 3.3-70B	Baseline	-9.19 $\pm$ 1.54	-0.091 $\pm$ 0.021	16.90 $\pm$ 0.82	30.28 $\pm$ 11.87	22.67 $\pm$ 8.39
	Reflection	-8.44 $\pm$ 1.58	-0.087 $\pm$ 0.025	16.36 $\pm$ 0.31	44.69 $\pm$ 13.25	27.67 $\pm$ 1.15
	Adaptive-OPRO	-6.16$\pm$2.08	-0.066$\pm$0.004	14.05$\pm$3.33	54.36$\pm$12.44	28.33 $\pm$ 3.21
Qwen3-235B	Baseline	-1.78 $\pm$ 3.86	-0.006 $\pm$ 0.039	13.09 $\pm$ 1.88	36.51 $\pm$ 17.55	13.00 $\pm$ 4.00
	Reflection	-5.76 $\pm$ 2.97	-0.049 $\pm$ 0.033	14.18 $\pm$ 1.91	25.00 $\pm$ 0.00	8.67 $\pm$ 0.58
	Adaptive-OPRO	1.33$\pm$1.91	0.025$\pm$0.019	11.41$\pm$0.06	50.00$\pm$0.00	9.00 $\pm$ 0.00
Qwen3-32B	Baseline	-10.62 $\pm$ 3.54	-0.087 $\pm$ 0.031	16.72 $\pm$ 2.75	30.00 $\pm$ 10.00	25.33 $\pm$ 1.53
	Reflection	-7.76 $\pm$ 0.90	-0.065 $\pm$ 0.002	16.47 $\pm$ 3.44	28.72 $\pm$ 25.06	31.67 $\pm$ 2.31
	Adaptive-OPRO	-3.48$\pm$2.19	-0.022$\pm$0.021	15.52$\pm$0.68	43.45$\pm$6.27	28.67 $\pm$ 1.53
Claude Sonnet 4	Baseline	-7.26 $\pm$ 2.99	-0.066 $\pm$ 0.030	17.59 $\pm$ 1.55	31.19 $\pm$ 7.84	13.00 $\pm$ 4.36
	Reflection	-5.69 $\pm$ 1.82	-0.058 $\pm$ 0.013	15.12 $\pm$ 3.26	46.67$\pm$5.77	12.67 $\pm$ 2.08
	Adaptive-OPRO	0.35$\pm$1.78	0.008$\pm$0.018	14.76$\pm$2.87	43.45 $\pm$ 6.27	15.00 $\pm$ 2.00
Claude Sonnet 4 w/ Thinking	Baseline	-4.46 $\pm$ 4.76	-0.043 $\pm$ 0.048	14.32 $\pm$ 4.12	11.11 $\pm$ 19.24	14.00 $\pm$ 2.65
	Reflection	-8.60 $\pm$ 0.59	-0.078 $\pm$ 0.004	19.45 $\pm$ 1.65	14.29 $\pm$ 24.75	11.67 $\pm$ 2.08
	Adaptive-OPRO	-0.73$\pm$3.82	-0.004$\pm$0.038	12.94$\pm$2.32	43.89$\pm$21.11	17.00 $\pm$ 5.00
GPT-o4-mini	Baseline	-1.30 $\pm$ 1.71	-0.017 $\pm$ 0.017	9.68$\pm$3.12	29.17 $\pm$ 11.02	15.33 $\pm$ 3.06
	Reflection	-2.52 $\pm$ 4.03	-0.039 $\pm$ 0.045	9.82 $\pm$ 3.43	51.28 $\pm$ 5.06	20.33 $\pm$ 3.06
	Adaptive-OPRO	9.06$\pm$0.73	0.094$\pm$0.008	11.48 $\pm$ 0.00	65.28$\pm$16.84	17.33 $\pm$ 5.86
GPT-o3	Baseline	-6.11 $\pm$ 3.42	-0.080 $\pm$ 0.029	11.58 $\pm$ 3.09	42.59 $\pm$ 8.49	18.67 $\pm$ 3.21
	Reflection	-4.60 $\pm$ 3.40	-0.053 $\pm$ 0.044	12.11 $\pm$ 1.27	46.03 $\pm$ 16.88	18.33 $\pm$ 2.52
	Adaptive-OPRO	9.02$\pm$3.28	0.146$\pm$0.048	5.33$\pm$0.14	72.81$\pm$17.27	19.67 $\pm$ 4.16

Table 1: Performance comparison between non-LLM-based and LLM-based approaches using ATLAS in volatile, declining market conditions. **Bold** values indicate the best per model.

goal is to isolate the *adaptation mechanism* under identical data and execution semantics. We therefore evaluate all methods within a single, transparent setup rather than re-implementing full external agent stacks, which differ in action spaces, state representations, and execution interfaces. We include reflection as a widely used and portable form of sequential feedback, providing a focused comparison to *Adaptive-OPRO* and the fixed baseline.

Non-LLM baselines. Following Li et al. (2024), we include five widely used quantitative strategies to contextualize results: Buy & Hold, MACD (Wang and Kim, 2018), SMA (Gencay, 1996), SLMA (Wang and Kim, 2018), and Bollinger Bands (Day et al., 2023). For window-based methods, we test multiple window lengths per regime and report a strong, representative configuration for each strategy (e.g., 10-day SMA; 10/30-day SLMA). Full specifications in App. D.8.

Execution environment. Agents interact with StockSim (Papadakis et al., 2025) via an *order-level* action space, requiring CTAs to submit fully *executable* orders (type, side, size, price). Compared to signal- or position-level formulations common in prior LLM trading studies (Li et al., 2024; Xiao et al., 2025), this enforces execution feasibility

(cash, inventory, validity) and yields a complete audit trail of orders, fills, and portfolio states. Consistent with standard offline evaluation, we abstract away four aspects of market microstructure, namely latency, slippage, market impact, and intraday partial fills, and assume deterministic execution. Since these abstractions apply uniformly to every prompting strategy, they shift absolute return levels while leaving comparisons between Baseline, Reflection, and Adaptive-OPRO internally consistent.

Evaluation Metrics. We employ five metrics capturing different aspects of trading performance:

Return on Investment (ROI): Total percentage return calculated as:

$$\frac{\text{final value} - \text{initial value}}{\text{initial value}} \times 100 \quad (2)$$

where portfolio values include both cash holdings and the current market value of all stocks owned.

Sharpe Ratio (SR): Risk-adjusted return metric:

$$\frac{\mu - r_f}{\sigma} \quad (3)$$

where μ is mean daily return, σ is daily return standard deviation, and r_f is the risk-free rate (set to 0 as in Li et al. (2024)).

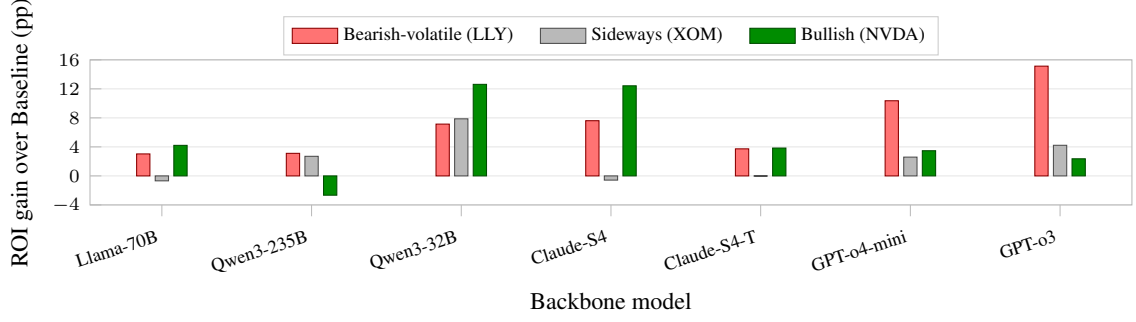


Figure 3: ROI gain of Adaptive-OPRO over the Baseline prompt, in percentage points, per backbone and market regime. The effect is most pronounced under bearish-volatile conditions.

Maximum Drawdown (DD): The worst peak-to-trough decline in portfolio value:

$$\max_{t \in [0, T]} \left(\max_{s \in [0, t]} V_s - V_t \right) / \max_{s \in [0, t]} V_s \quad (4)$$

where V_t is portfolio value at time t . This measures the largest loss from any historical high, reflecting downside risk and stress tolerance.

Win Rate: Percentage of *profitable closed* (i.e. completed) trades, computed as:

$$\frac{\text{Closed trades with realized profit} > 0}{\text{Total closed trades}} \times 100. \quad (5)$$

“Closed trades” are fully opened and exited positions; open positions are excluded. Win rate reflects decision consistency but does not ensure profitability if losses outweigh gains.

Number of Trades: Total trading frequency over the evaluation period. Higher frequencies indicate active, opportunistic short-term strategies, while lower frequencies suggest patient, conviction-driven approaches. Additional metrics, results, and analyses are reported in App. E.

6 Results

Tables 1, 2, and 3 present the results of our experimental design evaluating ATLAS across diverse market conditions. The results show that *Adaptive-OPRO* consistently improves upon fixed prompts across models and market conditions, while reflection often deteriorates performance or provides inconsistent value. Non-LLM strategies demonstrate regime-dependent performance, with different technical approaches succeeding in specific conditions but failing to generalize. ATLAS with *Adaptive-OPRO* delivers stable performance across tested regimes, with certain model pairings achieving positive returns even in volatile and declining market

conditions where most baseline strategies struggle. The order-level action space reveals distinct patterns across model families and supports attribution from analytical reasoning to execution behavior.

6.1 Optimization in Sequential Decision-Making

Adaptive-OPRO consistently outperforms both static baseline prompts and reflection-based approaches across the tested models and market conditions. Figure 3 makes this breadth explicit: the ROI gain over Baseline is positive across most backbones and regimes and spans all four model families, so the improvement is not confined to a particular architecture or market condition. Its magnitude scales with the slack left by the tuned baseline, which is largest under volatile conditions and narrowest in already-favorable regimes.

Return, risk-adjusted, and win-rate metrics jointly indicate successful adaptation to market feedback: models paired with *Adaptive-OPRO* achieve higher returns while maintaining or reducing drawdowns, with SR gains showing that improvements arise from strategic enhancement and not increased risk-taking. Crucially, these return gains are accompanied by higher win rates, indicating consistent decision-making rather than sporadic large profits masking frequent losses. For instance, in the bearish regime (Table 1), GPT-o3 and GPT-o4-mini shift from negative baseline returns to strong positive performance under *Adaptive-OPRO*, while Qwen3-235B moves from losses to gains. This pattern persists across range-bound and bullish conditions, suggesting that prompt optimization captures regime-appropriate cues rather than overfitting to specific market settings.

Comparisons to baseline performance. We examine how baseline decision quality relates to the gains from *Adaptive-OPRO* under volatile, declin-

Model	Prompting	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
Non-LLM-Based Strategies						
Buy & Hold	N/A	1.14	0.013	6.97	0.00	1
MACD	N/A	-0.26	-0.019	5.90	0.00	3
SMA	N/A	-0.13	-0.019	5.57	0.00	3
SLMA	N/A	-1.12	-0.043	5.28	0.00	2
Bollinger Bands	N/A	0.00	0.000	0.00	0.00	0
LLM-Based Strategies - ATLAS						
Llama 3.3-70B	Baseline	-0.42 ± 2.06	-0.024 ± 0.051	5.56 ± 1.08	53.48 ± 9.56	26.00 ± 2.00
	Reflection	-2.61 ± 0.77	-0.083 ± 0.014	6.38 ± 0.72	46.63 ± 3.15	26.33 ± 6.51
	Adaptive-OPRO	-1.10 ± 0.44	-0.045 ± 0.012	5.15 ± 0.71	50.00 ± 3.85	25.33 ± 1.15
Qwen3-235B	Baseline	-2.43 ± 0.68	-0.04 ± 0.01	5.72 ± 0.16	46.67 ± 5.77	11.66 ± 0.57
	Reflection	-2.02 ± 1.44	-0.04 ± 0.03	6.26 ± 1.77	36.51 ± 5.50	13.33 ± 2.31
	Adaptive-OPRO	0.27 ± 1.83	0.01 ± 0.04	7.20 ± 2.09	32.86 ± 15.45	11.00 ± 3.61
Qwen3-32B	Baseline	-9.14 ± 1.02	-0.20 ± 0.02	9.82 ± 0.90	28.85 ± 17.20	21.00 ± 1.73
	Reflection	-7.96 ± 3.11	-0.16 ± 0.06	9.05 ± 2.90	40.55 ± 15.48	24.33 ± 3.05
	Adaptive-OPRO	-1.27 ± 3.21	-0.03 ± 0.07	6.75 ± 0.54	35.83 ± 2.57	25.67 ± 5.50
Claude Sonnet 4	Baseline	-4.49 ± 4.22	-0.134 ± 0.114	7.71 ± 1.06	37.50 ± 4.17	19.00 ± 3.46
	Reflection	-3.78 ± 4.23	-0.115 ± 0.105	10.54 ± 1.58	23.84 ± 8.27	18.00 ± 6.93
	Adaptive-OPRO	-5.07 ± 4.53	-0.165 ± 0.143	9.23 ± 2.71	31.02 ± 7.90	18.33 ± 2.52
Claude Sonnet 4 w/ Thinking	Baseline	-0.99 ± 0.80	-0.039 ± 0.020	7.75 ± 1.00	56.28 ± 1.50	17.00 ± 5.20
	Reflection	-1.49 ± 3.76	-0.069 ± 0.123	7.27 ± 2.26	45.11 ± 12.60	17.00 ± 5.57
	Adaptive-OPRO	-1.01 ± 0.90	-0.046 ± 0.020	5.16 ± 0.52	36.20 ± 24.47	16.33 ± 2.08
GPT-o4-mini	Baseline	1.29 ± 1.38	0.021 ± 0.044	3.23 ± 0.48	39.01 ± 3.61	22.67 ± 7.57
	Reflection	-1.48 ± 0.54	-0.087 ± 0.018	4.64 ± 0.75	32.62 ± 7.49	27.33 ± 3.06
	Adaptive-OPRO	3.88 ± 2.21	0.089 ± 0.067	3.28 ± 0.95	47.95 ± 7.15	25.33 ± 5.03
GPT-o3	Baseline	-0.60 ± 1.71	-0.034 ± 0.050	5.93 ± 1.33	60.74 ± 5.59	16.33 ± 2.52
	Reflection	-1.55 ± 2.09	-0.084 ± 0.075	5.02 ± 0.72	42.50 ± 6.61	16.67 ± 0.58
	Adaptive-OPRO	3.62 ± 0.90	0.096 ± 0.027	3.46 ± 0.48	71.93 ± 15.90	16.00 ± 2.65

Table 2: Performance comparison between non-LLM-based and LLM-based approaches using ATLAS in range-bound market conditions. **Bold** values indicate the best results per model.

ing markets. Baseline and Adaptive-OPRO ROI are moderately correlated ($r = 0.64$), suggesting that *stronger baselines* maintain *higher absolute returns* after adaptation. However, the improvement over baseline shows no meaningful correlation ($r = 0.05$) and an almost flat gradient ($\beta \approx 0.06$). This indicates that Adaptive-OPRO does not simply amplify existing strengths, but delivers *improvements largely independent* of initial performance. Similar trends hold for risk-adjusted metrics, implying that Adaptive-OPRO mainly alters decision behavior rather than scaling baseline profitability.

The reflection paradox. In contrast, reflection-based prompting (Li et al., 2024) exhibits a markedly different behavior. In the bearish-volatile regime, the ROI improvement under reflection shows a strong negative correlation with baseline performance ($r = -0.78$, $p < 0.05$), accompanied by a pronounced negative performance gradient ($\beta = -0.61$), indicating that models with *stronger baseline* decision quality tend to *deteriorate more* with reflection. This suggests that reflection does not merely fail to improve performance, but can actively *disrupt effective decision policies* in high-noise environments. Rather than stabilizing behavior, reflection appears to **amplify stochasticity** and

override useful heuristics, particularly for models that already exhibit competent baseline trading strategies, consistent with overthinking induced by redundant information (more details in App. H).

6.2 Trading Behavior Across LLMs

The order-level action space reveals systematic behavioral differences across model families, with performance broadly correlating with general model capabilities. Beyond averages, variance across runs captures decision reliability, especially when timing and sizing errors are amplified.

GPT models exhibit distinct trading styles and adaptation patterns. **GPT-o3** integrates inputs from agents into coherent decisions, showing conservative risk management that can cap gains in strongly trending markets but delivers consistent performance across regimes. This manifests as robust returns with comparatively low drawdowns and low run-to-run variance, denoting stable execution.

GPT-o4-mini emphasizes short-term risk control through frequent stop-losses and early profit-taking. This behavior aligns with stronger outcomes in volatile settings and more muted trend capture in sustained moves; it also tends toward higher trading frequency in some regimes. Still,

Model	Prompting	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
Non-LLM-Based Strategies						
Buy & Hold	N/A	41.30	0.409	3.16	0.00	1
MACD	N/A	-0.62	-0.343	0.62	0.00	1
SMA	N/A	36.77	0.384	3.12	0.00	1
SLMA	N/A	15.88	0.254	2.98	0.00	1
Bollinger Bands	N/A	0.00	0.000	0.00	0.00	0
LLM-Based Strategies - ATLAS						
Llama 3.3-70B	Baseline	37.86 $\pm$ 12.31	0.388 $\pm$ 0.096	3.46 $\pm$ 0.63	20.37 $\pm$ 35.28	13.00 $\pm$ 20.78
	Reflection	40.40 $\pm$ 1.43	0.422$\pm$ 0.023	2.96$\pm$ 0.34	33.33 $\pm$ 57.74	5.33 $\pm$ 6.66
	Adaptive-OPRO	42.07$\pm$ 1.85	0.418 $\pm$ 0.016	3.15 $\pm$ 0.02	100.00$\pm$ 0.00	1.33 $\pm$ 0.58
Qwen3-235B	Baseline	43.91$\pm$ 2.31	0.42$\pm$ 0.00	3.34 $\pm$ 0.16	0.00 $\pm$ 0.00	2.00 $\pm$ 0.00
	Reflection	34.08 $\pm$ 12.30	0.37 $\pm$ 0.08	2.98$\pm$ 0.30	23.81$\pm$ 41.24	11.33 $\pm$ 16.17
	Adaptive-OPRO	41.25 $\pm$ 0.00	0.42$\pm$ 0.00	3.16 $\pm$ 0.00	0.00 $\pm$ 0.00	2.00 $\pm$ 0.00
Qwen3-32B	Baseline	35.75 $\pm$ 5.35	0.48$\pm$ 0.06	2.86$\pm$ 0.30	60.86 $\pm$ 52.71	22.33 $\pm$ 3.06
	Reflection	41.72 $\pm$ 1.32	0.43 $\pm$ 0.01	3.03 $\pm$ 0.22	66.67 $\pm$ 57.74	10.67 $\pm$ 5.13
	Adaptive-OPRO	48.37$\pm$ 0.10	0.47 $\pm$ 0.00	3.15 $\pm$ 0.02	100.00$\pm$ 0.00	18.00 $\pm$ 5.00
Claude Sonnet 4	Baseline	13.43 $\pm$ 8.62	0.180 $\pm$ 0.121	5.52 $\pm$ 3.96	60.83$\pm$ 12.30	21.67 $\pm$ 9.50
	Reflection	5.21 $\pm$ 1.10	0.089 $\pm$ 0.026	5.11 $\pm$ 1.86	39.25 $\pm$ 15.79	22.33 $\pm$ 1.53
	Adaptive-OPRO	25.85$\pm$ 10.61	0.290$\pm$ 0.087	3.75$\pm$ 0.59	43.81 $\pm$ 38.37	19.00 $\pm$ 12.17
Claude Sonnet 4 w/ Thinking	Baseline	12.52 $\pm$ 2.47	0.175 $\pm$ 0.030	5.03 $\pm$ 1.53	53.30 $\pm$ 14.47	17.00 $\pm$ 2.65
	Reflection	11.12 $\pm$ 4.86	0.186 $\pm$ 0.083	3.42$\pm$ 2.23	77.86$\pm$ 2.58	17.00 $\pm$ 5.00
	Adaptive-OPRO	16.36$\pm$ 7.87	0.217$\pm$ 0.105	5.18 $\pm$ 2.52	68.89 $\pm$ 30.06	12.67 $\pm$ 4.04
GPT-o4-mini	Baseline	7.00 $\pm$ 3.46	0.125 $\pm$ 0.054	2.74 $\pm$ 0.79	46.29 $\pm$ 3.21	18.67 $\pm$ 1.53
	Reflection	9.80 $\pm$ 3.21	0.189 $\pm$ 0.067	2.45$\pm$ 1.00	54.54 $\pm$ 7.92	26.33 $\pm$ 9.61
	Adaptive-OPRO	10.47$\pm$ 3.84	0.193$\pm$ 0.046	3.42 $\pm$ 0.90	62.70$\pm$ 11.25	20.33 $\pm$ 2.89
GPT-o3	Baseline	22.70 $\pm$ 0.92	0.269 $\pm$ 0.029	6.82 $\pm$ 3.03	66.67 $\pm$ 28.87	7.33 $\pm$ 2.52
	Reflection	21.98 $\pm$ 4.54	0.325 $\pm$ 0.040	3.14 $\pm$ 0.99	96.67 $\pm$ 5.77	18.00 $\pm$ 3.61
	Adaptive-OPRO	25.06$\pm$ 4.28	0.392$\pm$ 0.019	2.31$\pm$ 0.80	100.00$\pm$ 0.00	9.67 $\pm$ 4.04

Table 3: Performance comparison between non-LLM-based and LLM-based approaches using ATLAS in bullish market conditions. **Bold** values indicate the best per model.

Adaptive-OPRO generally improves its consistency and profitability relative to fixed prompting, with moderate variance suggesting a more reactive but still controlled policy.

Qwen models show divergent behavior based on scale. **Qwen3-235B** trades more selectively and, across several regimes, achieves stable positive outcomes under Adaptive-OPRO. All three tables reflect that prompt adaptation is important here: it often turns otherwise marginal/negative behavior into positive returns while keeping activity relatively restrained, consistent with risk-reward balancing.

Qwen3-32B is more active and variable, with larger swings across runs and regimes. Adaptive-OPRO improves its behavior, reducing losses in adverse settings and strengthening performance in favorable ones, but residual variance suggests less stable execution than the larger variant.

Llama 3.3-70B adopts simpler trading strategies with limited risk-management sophistication. Qualitatively, it shows delayed responses to market shifts and occasional abrupt changes in stance, which correspond to weaker performance in more adversarial regimes. Interestingly, this straightforward behavior performs well in the bullish regime in our results, consistent with capturing upward

drift without overcomplicating execution.

Claude Sonnet 4 varies depending on reasoning mode, with variance patterns revealing meaningful differences in reliability. Certain configurations exhibit markedly higher run-to-run variability, indicating less predictable decision-making. With extended thinking enabled, the model often produces detailed analysis but the results show mixed execution quality; without thinking, decisions become more erratic and consistency across regimes degrades, suggesting that the bottleneck is both analysis depth and subsequent order construction.

Overall, the key insight enabled by order-level specifications is that weaker configurations often generate plausible market analysis but fail in position sizing, timing, or order selection, whereas successful configurations consistently translate analysis into coherent execution.

6.3 LLM Optimization Capabilities

A key advantage of *Adaptive-OPRO* is that optimization yields *interpretable* instruction updates, assessed along two axes: (i) whether the revised prompt is objectively aligned with the trading goal (e.g., explicit risk controls, sizing discipline, and when to trade), and (ii) whether those instructions

Regime	Configuration	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
Bearish Regime	No News	4.07 $\pm$ 0.72	0.056 $\pm$ 0.016	7.84$\pm$ 3.15	53.51 $\pm$ 6.67	25.33 $\pm$ 4.51
	No Market Data	-5.75 $\pm$ 0.76	-0.094 $\pm$ 0.017	11.32 $\pm$ 2.63	37.52 $\pm$ 4.87	18.33 $\pm$ 3.06
	No News & No Market	-6.86 $\pm$ 1.68	-0.078 $\pm$ 0.036	14.54 $\pm$ 3.30	43.94 $\pm$ 6.94	22.33 $\pm$ 1.15
	ATLAS	9.06$\pm$ 0.73	0.094$\pm$ 0.008	11.48 $\pm$ 0.00	65.28$\pm$ 16.84	17.33 $\pm$ 5.86
Sideways Regime	No News	-8.20 $\pm$ 1.64	-0.264 $\pm$ 0.069	9.09 $\pm$ 2.99	22.82 $\pm$ 13.65	35.00 $\pm$ 12.29
	No Market Data	0.01 $\pm$ 0.92	-0.011 $\pm$ 0.021	6.56 $\pm$ 1.58	46.55 $\pm$ 23.15	13.33 $\pm$ 3.06
	No News & No Market	-4.60 $\pm$ 0.70	-0.136 $\pm$ 0.026	7.01 $\pm$ 2.29	35.26 $\pm$ 13.09	21.00 $\pm$ 4.58
	ATLAS	3.88$\pm$ 2.21	0.089$\pm$ 0.067	3.28$\pm$ 0.95	47.95$\pm$ 7.15	25.33 $\pm$ 5.03
Bullish Regime	No News	6.62 $\pm$ 0.25	0.090 $\pm$ 0.008	6.67 $\pm$ 0.36	41.96 $\pm$ 5.21	28.33 $\pm$ 4.62
	No Market Data	11.78$\pm$ 1.76	0.216$\pm$ 0.024	3.70 $\pm$ 0.86	70.24$\pm$ 14.03	20.00 $\pm$ 5.57
	No News & No Market	7.34 $\pm$ 2.79	0.110 $\pm$ 0.012	5.76 $\pm$ 2.01	63.84 $\pm$ 9.39	20.67 $\pm$ 1.53
	ATLAS	10.47 $\pm$ 3.84	0.193 $\pm$ 0.046	3.42$\pm$ 0.90	62.70 $\pm$ 11.25	20.33 $\pm$ 2.89

Table 4: Ablation study results showing individual agent contributions using GPT-o4-mini across three market regimes. **Bold** values indicate the best results per configuration.

are reflected in subsequent order-level behavior (frequency, timing, and position sizing). After manual inspection of the results, we observe clear family-level patterns. **GPT models** consistently produce well-structured, objective-aligned refinements that translate observed weaknesses into actionable constraints, and these updates tend to be followed in execution, consistent with their lower run-to-run variance. **Qwen models** also generate targeted improvements, with the larger Qwen3-235B producing more coherent and internally consistent instruction revisions, which aligns with its more stable selective trading behavior. In contrast, **Llama** often reports edits that are not present in the actual prompt or proposes changes that conflict with the stated objective, weakening the connection between optimization output and downstream execution. **Claude models** often shift toward increasingly procedural and restrictive prompts, which can reduce adaptability; notably, this prescriptiveness does not reliably translate into stable execution, as reflected by higher variance in several configurations. Examples of observed patterns in App. G.

6.4 ATLAS Ablation Study

Table 4 shows distinct agent contributions through performance drops when each is ablated.

Market Analyst is a core component across market regimes. Its removal consistently results in the most significant performance degradation, especially in challenging conditions such as the bearish-volatile regime, where technical context is crucial for decision-making. In the sideways regime, the absence of market analysis not only reduces returns but also lowers trading frequency, suggesting that agents lose confidence to act without a solid technical foundation. Notably, in the bullish regime, ROI slightly improves when market data is excluded,

suggesting that in up-trending conditions social consensus may offer cleaner entry signals.

News analyst contributes regime-specific strategic value. In the bullish regime, news removal leads to lower returns as agents become more conservative. The sideways regime shows news analysis as critical, with its removal producing severe degradation, suggesting that sentiment analysis is essential when technical signals are ambiguous.

Combination of News & Market Analyst highlights the complementary value of these signals. Across all regimes, removing both agents substantially degrades performance, showing that news and market data provide non-redundant information. In the bearish regime, the drop reflects the importance of sentiment and technical context under volatility, while in the sideways regime their absence produces unstable, unprofitable behavior. Even in the bullish regime, combined removal harms performance, indicating that each component contributes differently across regimes and that their joint effect is not simply additive.

7 Conclusion

In this work, we introduce ATLAS, an LLM-based trading framework that combines *Adaptive-OPRO* for prompt optimization under delayed, noisy feedback with structured analyst inputs and an order-level interface. Across regimes and model families, Adaptive-OPRO outperforms tuned static prompts, while standard reflection proves inconsistent. The order-level interface reveals model-specific trading behaviors and separates analytical quality from execution choices, enabling clearer attribution and interpretability. ATLAS with Adaptive-OPRO provides a practical, reliable, auditable paradigm for sequential LLM decision-making.

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Limitations

Following prior LLM-agent and market-simulation work, we focus on three liquid equities over two-month, regime-specific windows with daily decisions to reduce confounding from asset heterogeneity and shifting market structure. This isolates adaptation effects under a shared interface but does not support generalization across assets, sectors, horizons, or macro conditions. Results should be read as behavioral evidence about *Adaptive-OPRO*, not as market-wide performance claims.

End-of-day decisions provide stable feedback for optimization under delayed, noisy outcomes, but prevent agents from reacting to intraday moves or capturing timing-dependent behaviors. The frictionless execution assumption discussed in Section 5 also means that strategies sensitive to fill quality, such as high-turnover or large-size policies, would need revalidation under a microstructure-aware simulator before any live-trading claim.

Each configuration runs three times due to resource constraints, capturing stochastic variance but limiting statistical power. Comparisons isolate prompt adaptation under a shared order-level interface rather than varying full system architectures. While order-level actions improve interpretability by separating analysis from execution, we do not include a directional-only ablation for direct causal comparison. Finally, although we cover multiple model families (GPT, Claude, Llama, Qwen), behaviors may vary with architectures, scales, and training procedures beyond those studied here.

Ethical Considerations

This work focuses on controlled, simulated trading experiments to study prompt optimization and does not involve real-world financial transactions or human subjects. All analyses are conducted in a reproducible, transparent environment, minimizing potential risks. While findings provide insights into model behavior, they are not financial advice and should not be used for live trading.

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A Financial Markets and Trading Foundations

This appendix summarizes the trading concepts needed to interpret an *order-aware* interface and the signals used by the MARKET ANALYST. The focus is on how ATLAS expresses decisions as executable orders in *StockSim* rather than on venue-specific microstructure.

A.1 Orders and Positions

ATLAS expresses actions at the order level and supports both long and short positioning.

Order types. **Market orders** seek immediate execution at the best available prices and prioritize certainty of fill over price control. **Limit orders** specify a worst acceptable price for buys or a best acceptable price for sells and prioritize price control over certainty of execution. **Stop orders** activate once a trigger is reached and are commonly used for risk control or momentum entry.

Long and short. A **buy to open** creates or increases a long position. A **sell short** creates a short position that profits if price declines. Exits are expressed symmetrically as **sell to close** for long positions and **buy to cover** for short positions. The Central Trading Agent may attach stops or limits to manage risk and profit-taking for either side.

Decision cadence. The Central Trading Agent makes decisions on a daily schedule. At each decision point it consumes the updated analyst summaries and current portfolio state, then may submit new or modifying orders that are evaluated by *StockSim* under standard semantics. At initialization, the portfolio holds \$100,000 in cash and no positions. Since our headline metrics are percentage based (e.g., ROI, Sharpe, and drawdown computed from returns), the absolute starting capital does not affect reported performance and only scales dollar P&L.

A.2 Regime Taxonomy

We organize evaluation windows by broad market regimes in order to study behavior under distinct conditions.

Bearish volatile denotes periods with sustained downward drift and elevated variability. **Sideways** denotes range-bound behavior with mixed signals and limited trend persistence. **Bullish** denotes periods with sustained upward drift and comparatively

orderly pullbacks. In the main experiments we instantiate one window for each regime and keep the decision cadence and interface fixed. The taxonomy is agnostic to any single indicator choice and can be operationalized by simple trend and volatility summaries when needed.

B Technical Indicators Used in Market Analysis

This appendix provides detailed explanations of the technical indicators employed by the Market Analyst agent in ATLAS, covering their mathematical formulations, implementation specifics, and interpretive significance in financial market analysis. All technical indicators described in this section are calculated by the *StockSim* (Papadakis et al., 2025) simulation environment and integrated into our analysis framework to provide comprehensive market insights.

Data source. The Market Analyst consumes OHLCV, volume, and session VWAP series from Massive¹ for the specified instrument and evaluation window. Bars are retrieved at daily resolution and aligned to official U.S. market sessions, with corporate actions (splits and dividends) from Massive used to adjust prices consistently with *StockSim*. All technical indicators described in this appendix are computed inside *StockSim* from these Massive-derived bars. Days with incomplete or missing bars are excluded rather than backfilled, and no survivorship or lookahead adjustments are applied beyond standard split and dividend handling.

B.1 Simple Moving Average (SMA) and Exponential Moving Average (EMA)

Simple Moving Average (SMA): The SMA is calculated as the arithmetic mean of closing prices over a specified number of periods (Murphy, 1999):

$$SMA_n = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \quad (6)$$

where P_t represents the closing price at time t and n is the number of periods. For our analysis, we employ SMA periods of 20, 50, 100, and 200 days to capture short-term, medium-term, and long-term trend characteristics. SMA provides equal weight to all prices in the calculation period, which makes it suitable for identifying longer-term trends but

¹<https://massive.com>

less responsive to recent price changes (Murphy, 1999).

Exponential Moving Average (EMA): The EMA assigns exponentially decreasing weights to older prices, which makes it more responsive to recent price movement (Murphy, 1999):

$$EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1} \quad (7)$$

where $\alpha = \frac{2}{n+1}$ is the smoothing factor and n is the number of periods. In our implementation, we utilize 12-period and 26-period EMAs, which serve as the foundation for MACD calculation and provide complementary trend analysis to our SMA suite. Research indicates that EMA often outperforms SMA in volatile conditions due to its enhanced sensitivity to recent price changes (Kaufman, 2013).

B.2 Relative Strength Index (RSI)

The RSI is a momentum oscillator that measures the speed and magnitude of price changes, oscillating between 0 and 100 (Wilder, 1978):

$$RSI = 100 - \frac{100}{1 + RS} \quad (8)$$

where $RS = \frac{\text{Average Gain}}{\text{Average Loss}}$ over a specified period. Our analysis uses the standard 14-day period as originally recommended by Wilder (1978). The average gain and loss are calculated using exponential smoothing as originally formulated:

$$\bar{G}_t = \frac{13\bar{G}_{t-1} + G_t}{14} \quad (9)$$

$$\bar{L}_t = \frac{13\bar{L}_{t-1} + L_t}{14} \quad (10)$$

where $\bar{G}_t$ represents the average gain at time t , $\bar{L}_t$ represents the average loss at time t , G_t is the current gain, and L_t is the current loss. RSI values above 70 typically indicate overbought conditions, while values below 30 suggest oversold conditions (Wilder, 1978). These thresholds can be adapted to asset volatility and regime (Murphy, 1999).

B.3 Moving Average Convergence Divergence (MACD)

MACD is a trend-following momentum indicator that shows the relationship between two moving averages of a security's price (Murphy, 1999):

$$MACD = EMA_{12} - EMA_{26} \quad (11)$$

$$Signal\ Line = EMA_9(MACD) \quad (12)$$

$$Histogram = MACD - Signal\ Line \quad (13)$$

We employ the standard configuration. Crossovers and divergences are commonly used to identify trend changes and momentum shifts (Achelis, 2000).

B.4 Average True Range (ATR)

ATR measures market volatility by calculating the average of true ranges over a specified number of periods, as developed by Wilder (1978):

$$True\ Range = \max[(High - Low), |High - Close_{prev}|, |Low - Close_{prev}|] \quad (14)$$

$$ATR_n = \frac{1}{n} \sum_{i=0}^{n-1} TR_{t-i} \quad (15)$$

We use the standard 14-period ATR. ATR supports volatility-aware sizing and stop placement.

B.5 Bollinger Bands

Bollinger Bands consist of three lines: a middle band and two outer bands positioned at standard deviations above and below the middle band (Achelis, 2000):

$$\text{Middle Band} = SMA_{20} \quad (16)$$

$$\text{Upper Band} = SMA_{20} + (k \times \sigma) \quad (17)$$

$$\text{Lower Band} = SMA_{20} - (k \times \sigma) \quad (18)$$

where k is typically 2 and σ is the rolling standard deviation of close. The bands adapt to changing volatility and help contextualize extremes (Murphy, 1999).

B.6 Support and Resistance Levels

Support and resistance levels are price zones where the asset has historically shown difficulty moving below (support) or above (resistance) (Murphy, 1999). We focus on **horizontal** levels identified by repeated interactions and elevated volume. Their strength increases with the number of tests, traded volume, and time span.

B.7 Volume Profile

Volume Profile displays trading activity over price for a chosen window:

- **Point of Control (POC):** price with the highest traded volume
- **Value Area:** price range that contains a specified share of volume, typically 70%

- **High Volume Nodes:** locally elevated volume levels

Volume-based context helps identify zones where participation has been concentrated, which often align with support or resistance.

C Analyst Details

C.1 News Analyst

The *News Analyst* distills market-relevant information from financial news streams for a given ticker. Inputs are retrieved from the Massive API² as batches of timestamped items containing title, URL, summary, and keywords. The component produces a structured analysis along four dimensions that are stable across models and assets: *Sentiment Assessment*, *Key Developments*, *Market Relevance*, and *Source Analysis*. When headline-only context is insufficient, the analyst can fetch the full article text through an internal fetcher to improve coverage and reduce headline bias. The output is designed to be compact, auditable, and directly consumable by the Central Trading Agent; it does not generate trading signals.

Example input batch (NVDA).

##NEWS BATCH

[2025-04-28T12:45:00+00:00] Want to Avoid the “Magnificent Seven” and Generate Passive Income? This Vanguard ETF May Be for You — The Motley Fool

URL: <https://www.fool.com/investing/2025/04/28/magnificent-seven-passive-income-vanguard-etf/?source=iedfolrf0000001>

Summary: The article discusses the Vanguard High Dividend Yield ETF (VYM) as an alternative to large-cap tech, highlighting sector diversification and dividend income.

Keywords: Vanguard High Dividend Yield ETF, Magnificent Seven, passive income, value stocks, dividend stocks

[2025-04-28T07:15:00+00:00] This Could Be a Make-It-or-Break-It Moment for Nvidia. Here's What to Watch For This Week. — The Motley Fool

URL: <https://www.fool.com/investing/2025/04/28/a-make-it-or-break-it-moment-for-nvidia/?source=iedfolrf0000001>

Summary: Nvidia's near-term outlook hinges on AI demand signals from large customers reporting earnings this week.

Keywords: Nvidia, AI, Microsoft, Amazon, Meta Platforms, Alphabet

[2025-04-27T22:25:00+00:00] Prediction: Broadcom Stock Could Surge by 100% in the Next 3 Years — The Motley Fool

URL: <https://www.fool.com/investing/2025/04/27/prediction-broadcom-could-surge-by-100-in-the-next/?source=iedfolrf0000001>

Summary: Broadcom's networking and ASIC businesses are positioned to benefit from AI infrastructure demand.

Keywords: Broadcom, AI chips, networking, ASIC, VMware

²<https://massive.com>

Structured output (NVDA).

Sentiment Assessment Tone is cautiously optimistic and contingent on confirmation of continued AI spend by key customers. There is also a modest rotation narrative toward dividend payers that could weigh on high-beta tech.

Key Developments Earnings this week from Microsoft, Amazon, and Meta will signal near-term AI budgets. Alphabet's reaffirmed capital spending supports sustained demand for next-gen chips. Policy risks around import tariffs remain a headwind.

Market Relevance Expect higher volatility around customer earnings. Rotation into dividend ETFs could pressure large-cap AI beneficiaries if macro uncertainty rises. Broadcom's bullish outlook may drive sector flow dispersion that competes with Nvidia exposure.

Source Analysis All items originate from The Motley Fool, a retail-focused outlet with a constructive bias. Claims should be cross-checked against primary earnings releases and sell-side notes for actionable conviction.

Additional example (XOM).

Sentiment Assessment Mixed. ExxonMobil appears on a list of top buys for diversification strength, offset by policy uncertainty related to funding cuts for carbon capture projects.

Key Developments Federal funding for a \$332M CCS project at Baytown is being withdrawn, which may delay low-carbon hydrogen and ammonia plans, although core growth strategy remains intact.

Market Relevance Near-term noise in decarbonization headlines with limited change to base cash-flow trajectory. Integrated model and commercial partnerships support resilience.

Source Analysis Coverage from The Motley Fool blends stock-picking commentary with policy reporting and lacks direct primary citations. Verification from official releases is recommended when trading on policy moves.

Operational notes. The News Analyst refreshes daily in sync with the decision cadence, deduplicates near-identical headlines, and preserves a consistent schema across assets and regimes. Its role is to surface catalysts, stance shifts, and source reliability in a compact form that supports downstream reasoning by the Central Trading Agent.

C.2 Fundamental Analyst

The *Fundamental Analyst* extracts trading-relevant structure from periodic corporate disclosures (earnings releases, financial statements) and corporate actions (dividends, splits). It runs at low frequency to mirror real reporting cadence, typically activating once or twice per evaluation window. Inputs are retrieved via Massive³ and normalized to a compact schema consumed by the Central Trading Agent. The module does not emit buy/sell signals; it summarizes material changes and likely catalysts.

³<https://massive.com>

C.2.1 Financial Statement Components and Terminology

Revenue and income metrics.

- **Revenue** (net sales) is top-line activity prior to costs (Penman, 2012).

- **Gross profit margin:**

$$\text{GPM} = \frac{\text{Revenue} - \text{COGS}}{\text{Revenue}} \times 100\%, \quad (19)$$

capturing production efficiency and pricing power (Palepu et al., 2019).

- **Operating margin:**

$$\text{OpM} = \frac{\text{Operating Income}}{\text{Revenue}} \times 100\%, \quad (20)$$

reflecting core cost discipline (Penman, 2012).

- **Net income** is profit after all expenses, taxes, and interest.

- **Earnings per share (EPS):**

$$\text{EPS} = \frac{\text{Net Income}}{\text{Weighted Avg. Shares}}, \quad (21)$$

a per-share profitability anchor for valuation (Damodaran, 2012).

Cash-flow dynamics.

- **Operating cash flow (OCF)** approximates cash generated by operations:

$$\text{OCF} = \text{NI} + \text{NCE} \pm \text{WCC}, \quad (22)$$

where NI is net income, NCE non-cash expenses, WCC working-capital change (Penman, 2012).

- **Net cash flow** aggregates operating, investing, and financing cash flows:

$$\text{NCF} = \text{OCF} + \text{ICF} + \text{FCF}. \quad (23)$$

- **Capital allocation** covers capex, buybacks, dividends, and debt paydown, each with distinct market implications.

Balance-sheet metrics.

- **Total assets** and **total equity** summarize scale and residual value (Palepu et al., 2019).
- **Debt-to-equity** gauges leverage and risk:

$$\text{D/E} = \frac{\text{Total Debt}}{\text{Total Equity}}. \quad (24)$$

Higher values imply greater financial risk (Damodaran, 2012).

C.2.2 Corporate Actions and Structural Events

Stock splits. Splits increase share count while proportionally reducing price (e.g., 1:2, 1:4, 1:10), often to improve perceived affordability and liquidity (Baker and Powell, 1993).

Dividends.

- **Cash dividends** return capital to shareholders; policy signals management's view on reinvestment vs. distribution (Brealey et al., 2019).

- **Dividend yield:**

$$\text{Yield} = \frac{\text{Annual Dividends Per Share}}{\text{Current Price}} \times 100\%. \quad (25)$$

C.2.3 Analytical Dimensions

The analyst produces a concise, four-part summary focused on trading relevance: *Profit & Margin Trends*, *Cash Flow & Capital Allocation*, *Balance Sheet & Leverage / Earnings Quality flags*, and *Catalyst Watch*. Outputs are kept compact and directly auditable.

Example input batch (NVDA).

Stock Splits:

2024-06-10: 1:10 2021-07-20: 1:4 2007-09-11: 2:3
2006-04-07: 1:2

Dividends:

2025-03-12: \$0.010 2024-12-05: \$0.010 2024-09-12: \$0.010
2024-06-11: \$0.010 2024-03-05: \$0.040

Annual FY2025 (Filed: 2025-02-26):

Revenue \$130.5B; GPM 75.0%; OpM 62.4%; Net income \$72.9B; EPS \$2.94;
OCF \$64.1B; NCF \$1.3B; Assets \$111.6B; Equity \$79.3B; D/E 0.11.

Quarterly Q1 2025 (Filed: 2024-05-29):

Revenue \$26.0B; GPM 78.4%; OpM 64.9%; Net income \$14.9B; EPS \$5.98; NCF \$0.3B.

Quarterly Q2 2025 (Filed: 2024-08-28):

Revenue \$30.0B; GPM 75.1%; OpM 62.1%; Net income \$16.6B; EPS \$0.67; NCF \$1.0B.

Structured output (NVDA).

Profit & Margin Trends Q1→Q2 revenue grew ~15% to \$30B on sustained AI demand; gross margin held near 75% while operating margin eased from 64.9% to 62.1%, consistent with mix normalization.

Cash Flow & Capital Allocation FY25 OCF \$64B (~49% of sales) supports heavy capex and buybacks; net cash still positive. The cut in quarterly dividend from \$0.04 to \$0.01 signals prioritization of reinvestment.

Balance Sheet & Earnings Quality Low leverage and strong equity base support flexibility. The sharp EPS swing (Q1 \$5.98 vs. Q2 \$0.67) warrants a GAAP vs. non-GAAP review to isolate one-offs.

Catalyst Watch Upcoming guidance on AI trajectory, capex cadence, and inventory dynamics are potential volatility catalysts relative to consensus.

Additional example (XOM).

Profit & Margin Trends FY2024 net margin near 10% with operating margin $\sim 14\text{--}15\%$; quarterly prints show stability.

Cash Flow & Capital Allocation Strong free cash flow capacity; negative annual net cash reflects investing and distribution outflows (capex, buybacks, dividends) rather than operating stress.

Balance Sheet & Leverage Debt-free posture and current ratio > 1.3 provide high financial flexibility; equity base expanded through FY/Q3.

Catalyst Watch Capital-return actions (buyback/dividend changes) and updates on large projects are the near-term fundamental triggers.

D Experiments

D.1 Experimental Setup

Market regimes in our evaluation are instantiated using highly liquid, publicly traded equities selected prior to experimentation based on transparent criteria. Specifically, assets are required to exhibit stable liquidity conditions, clearly identifiable regime-consistent price dynamics over the evaluation window, and minimal microstructure distortions. This ensures that observed agent behavior reflects regime characteristics rather than artifacts of illiquidity or asset-specific noise. Asset instantiations are chosen independently of model performance and without outcome-driven adjustment, with selection criteria emphasizing representativeness of regime dynamics and sectoral diversity to reduce the likelihood that results are driven by idiosyncratic company- or industry-level effects.

Concretely, the bearish-volatile regime is instantiated using Eli Lilly and Company (LLY), the sideways regime using Exxon Mobil Corporation (XOM), and the bullish regime using NVIDIA Corporation (NVDA). All assets are evaluated over the same fixed two-month window (Apr 28–Jun 28, 2025) with a daily decision interval, ensuring consistency in sequential decision-making across regimes.

Importantly, ATLAS is asset- and regime-agnostic by design: no asset-specific features or regime-dependent assumptions are encoded in the framework, and the same experimental protocol can be directly applied to alternative equities, broader asset sets, or different evaluation horizons without modification.

D.2 Evaluation Scope

We evaluate ATLAS over a two-month window (28 Apr–28 Jun 2025) across three sector-diverse equities. This horizon provides multiple decision

cycles per asset while keeping full conversation histories within context limits and avoiding regime mixing. The period naturally includes routine corporate events and news, yielding a representative test bed.

D.3 Asset Selection Strategy

We use three equities chosen ex ante by simple, transparent criteria (liquidity, sector diversity, characteristic behavior): **NVDA** (technology, trending), **LLY** (healthcare, volatile drawdowns), **XOM** (energy, range-bound). This mix stresses different information channels and trading behaviors (trend capture, volatility management, and patience) without relying on outcome-driven selection.

D.4 Framework Configurations

Beyond the main-paper comparisons, we implemented additional variants to probe design choices:

- **Baseline:** Multi-agent with carefully engineered static prompts.
- **Adaptive-OPRO:** Prompt optimization applied only to the Central Trading Agent.
- **Reflection:** A reviewer agent that produces periodic feedback on recent decisions. We tested weekly reflections (as in prior work) and a shorter 1-day variant; the latter is exploratory and omitted from the main tables.
- **Adaptive-OPRO + Reflection:** Combined for interaction analysis; included here for completeness.

All runs keep analyst prompts fixed to isolate the adaptation mechanism at the decision layer.

D.5 Model Selection

We study how backbone capabilities translate to sequential decisions under identical interfaces:

- **Reasoning-enabled:** GPT-o3, GPT-o4-mini, Claude Sonnet 4 (thinking).
- **Matched base model:** Claude Sonnet 4 (no thinking) to isolate the effect of explicit reasoning.
- **Open-source:** Llama 3.3-70B, Qwen3-235B, Qwen3-32B to gauge transfer across families and deployment options.

Within a run, the same backbone powers all ATLAS components to avoid cross-model confounds.

D.6 Ablation Study Choices

To quantify information value within ATLAS, we run ablations exclusively under **GPT-o4-mini + Adaptive-OPRO**:

1. **No Market Analyst**: removes multi-timescale technical structure and indicators.
2. **No News Analyst**: removes unstructured text processing of headlines and stories.
3. **No Market & No News**: leaves only portfolio state and fundamentals.

We do not ablate the *Fundamental Analyst* due to its intentionally low activation frequency within these windows; its role is assessed qualitatively around reporting events. Each ablation is run three times.

D.7 Evaluation Methodology

We use a **multi-run protocol** of three independent runs per configuration and report mean $\pm$ standard deviation. Metrics mirror the main paper (returns, risk-adjusted returns, drawdowns, win rate on closed trades, and activity). In addition to aggregate metrics, we examine decision patterns and adaptation trajectories to explain *why* configurations differ.

D.8 Non-LLM Based Strategies

We compare against established trading strategies (Buy & Hold, moving average crossovers, MACD) that require no machine learning. These baselines contextualize LLM performance-showing where adds value versus simpler alternatives. A detailed description of these methods is presented below.

Buy and Hold The Buy and Hold strategy is a passive investment approach in which an asset is acquired at the beginning of the investment horizon and retained without any further trading actions, regardless of interim price fluctuations. This method assumes that, over time, the market tends to grow, and thus long-term holding can yield positive returns. It does not rely on any predictive model or technical indicator. In our evaluation, Buy and Hold serves as a benchmark strategy against which the performance of all other trading methods is compared.

Simple Moving Average (SMA) The SMA strategy (Gencay, 1996) issues trading signals based on the relationship between the current price of an

asset and its moving average over a fixed time window. Specifically, a buy (sell) signal is triggered when the price crosses above (below) the SMA. We test various window lengths selecting the optimal period based on validation performance.

Short-Long Moving Average (SLMA) The SLMA method (Wang and Kim, 2018) extends the SMA approach by employing two SMAs of different lengths: one short-term and one long-term. A buy signal is generated when the short-term average crosses above the long-term average, while a sell signal occurs at the inverse crossover.

Moving Average Convergence Divergence (MACD) The MACD strategy (Wang and Kim, 2018) captures momentum shifts by computing the difference between the 12-day and 26-day exponential moving averages. A 9-day EMA of the MACD line is used as a signal line. Trading signals are generated when the MACD line crosses the signal line from below (buy) or from above (sell). The exponential formulation ensures increased sensitivity to recent price movements.

Bollinger Bands The Bollinger Bands strategy (Day et al., 2023) incorporates volatility by constructing a band around a 20-day SMA, with the upper and lower bands placed two standard deviations above and below the mean, respectively. A price crossing above the upper band may indicate overbought conditions (sell signal), while crossing below the lower band may suggest oversold conditions (buy signal). We adopt the standard parameterization of 20-day SMA and multiplier 2, as commonly suggested in the literature.

Model	Prompting	Ann. SR $\uparrow$	Sortino $\uparrow$	ROIC (%) $\uparrow$	P/T (\$) $\uparrow$
LLM-Based Strategies - ATLAS					
Llama 3.3-70B	Baseline	6.16 $\pm$ 1.52	0.97 $\pm$ 0.22	30.98 $\pm$ 26.06	456.27 $\pm$ 790.29
	Reflection	6.70$\pm$ 0.37	1.03 $\pm$ 0.02	29.14 $\pm$ 21.06	1511.32$\pm$ 2617.69
	Adaptive-OPRO	6.63 $\pm$ 0.25	1.05$\pm$ 0.01	42.26$\pm$ 1.68	0.00
Qwen3-235B	Baseline	6.61 $\pm$ 0.02	0.67$\pm$ 0.00	40.90 $\pm$ 0.35	0.00 $\pm$ 0.00
	Reflection	5.94 $\pm$ 1.20	0.58 $\pm$ 0.14	27.90 $\pm$ 23.15	491.18$\pm$ 850.75
	Adaptive-OPRO	6.63$\pm$ 0.00	0.67$\pm$ 0.00	41.26$\pm$ 0.00	0.00 $\pm$ 0.00
Qwen3-32B	Baseline	7.57$\pm$ 0.96	0.63 $\pm$ 0.07	16.37 $\pm$ 21.12	1567.18 $\pm$ 1369.31
	Reflection	6.85 $\pm$ 0.18	0.67 $\pm$ 0.00	26.67 $\pm$ 17.99	3266.26$\pm$ 5812.42
	Adaptive-OPRO	7.41 $\pm$ 0.05	0.72$\pm$ 0.01	43.27$\pm$ 4.61	248.26 $\pm$ 200.41
Claude Sonnet 4	Baseline	2.86 $\pm$ 1.93	0.45 $\pm$ 0.33	2.82 $\pm$ 2.60	1212.88$\pm$ 920.24
	Reflection	1.42 $\pm$ 0.41	0.16 $\pm$ 0.05	0.86 $\pm$ 0.36	416.79 $\pm$ 149.76
	Adaptive-OPRO	4.60$\pm$ 1.38	0.68$\pm$ 0.22	8.25$\pm$ 9.83	371.70 $\pm$ 1779.64
Claude Sonnet 4 w/ Thinking	Baseline	2.78 $\pm$ 0.48	0.46 $\pm$ 0.20	3.27 $\pm$ 1.51	1246.39 $\pm$ 143.77
	Reflection	2.95 $\pm$ 1.32	0.57 $\pm$ 0.40	4.33 $\pm$ 1.72	1042.20 $\pm$ 424.00
	Adaptive-OPRO	3.45$\pm$ 1.66	0.76$\pm$ 0.56	5.44$\pm$ 2.81	2402.02$\pm$ 1239.52
GPT-o4-mini	Baseline	1.98 $\pm$ 0.86	0.27 $\pm$ 0.14	0.81 $\pm$ 0.39	212.27 $\pm$ 421.02
	Reflection	3.00 $\pm$ 1.06	0.47$\pm$ 0.23	1.40 $\pm$ 0.70	537.97$\pm$ 45.35
	Adaptive-OPRO	3.07$\pm$ 0.73	0.41 $\pm$ 0.12	1.54$\pm$ 0.47	506.75 $\pm$ 329.55
GPT-o3	Baseline	4.27 $\pm$ 0.47	0.61 $\pm$ 0.14	8.03 $\pm$ 1.86	4262.67$\pm$ 897.79
	Reflection	5.16 $\pm$ 0.63	0.68 $\pm$ 0.20	6.76 $\pm$ 2.76	2192.28 $\pm$ 920.54
	Adaptive-OPRO	6.22$\pm$ 0.30	1.22$\pm$ 0.37	17.04$\pm$ 7.65	3761.99 $\pm$ 749.07

Table 5: Additional performance metrics for NVDA (technology sector) comparing LLM-based approaches using ATLAS in bullish market conditions. Ann. SR = Annualized Sharpe Ratio, ROIC = Return on Invested Capital, P/T = Profit per Trade. **Bold** values indicate the best per model.

E Extended Results

This appendix consolidates additional metrics and analysis that complement the main paper’s results and experimental setup. All computations use *daily* portfolio returns with risk-free rate $r_f = 0$ and are reported as mean $\pm$ standard deviation over three independent runs, consistent with the protocol described in Section 5.

E.1 Additional Quantitative Results

Additional Evaluation Metrics. Beyond ROI, Sharpe Ratio, Maximum Drawdown, Win Rate, and Number of Trades, we report the following complementary measures:

Annualized Sharpe Ratio (Ann. SR):

$$\text{Ann. SR} = \text{SR} \times \sqrt{252},$$

which standardizes risk-adjusted performance to a yearly scale.

Sortino Ratio:

$$\text{Sortino} = \frac{\mu - r_f}{\sigma_d},$$

where μ is the mean daily return and σ_d is the standard deviation of negative daily returns only. This isolates downside variability.

Return on Invested Capital (ROIC):

$$\text{ROIC} = \frac{V_{\text{final}} - V_{\text{initial}}}{\sum_{i \in \text{entries}} p_i \cdot q_i} \times 100,$$

where V denotes portfolio value, p_i and q_i are the price and quantity of each opening trade. This measures net profit as a percentage of total capital committed to opening positions, capturing how efficiently the agent converts invested capital into returns.

Profit per Trade (P/T):

$$\text{P/T} = \frac{\text{Total net profit}}{\text{Number of trades}},$$

computed on *closed* round trips only. This reflects average value creation per completed decision cycle and should be interpreted alongside position-level outcomes and exposure management.

E.2 Risk-Adjusted Performance Validation

Extended risk-adjusted metrics reinforce the central findings (Tables 5, 6, 7). **Sortino Ratio** improvements under *Adaptive-OPRO* indicate that gains are not driven by larger risk-taking but by better mitigation of downside variability. The effect is strongest in the bearish-volatile regime, where lower downside dispersion coincides with tighter drawdown control. **ROIC** consistently rises with *Adaptive-OPRO* across model families, showing that optimization improves the efficiency of capital deployment rather than merely increasing turnover. Improvements in **P/T**, when paired with higher win rates, suggest more consistent decision quality and cleaner trade selection. Since P/T excludes open

Model	Prompting	Ann. SR $\uparrow$	Sortino $\uparrow$	ROIC (%) $\uparrow$	P/T (\$) $\uparrow$
LLM-Based Strategies - ATLAS					
Llama 3.3-70B	Baseline	-0.38 ± 0.81	-0.02 ± 0.06	-0.03 ± 0.16	-26.23 ± 164.36
	Reflection	-1.32 ± 0.21	-0.10 ± 0.01	-0.21 ± 0.07	-227.29 ± 38.58
	Adaptive-OPRO	-0.72 ± 0.19	-0.06 ± 0.02	-0.09 ± 0.03	-86.11 ± 31.28
Qwen3-235B	Baseline	-0.70 ± 0.22	-0.03 ± 0.01	-0.43 ± 0.13	-437.32 ± 151.36
	Reflection	-0.59 ± 0.54	-0.03 ± 0.02	-0.34 ± 0.25	-334.07 ± 245.20
	Adaptive-OPRO	0.17 ± 0.59	0.01 ± 0.03	-0.02 ± 0.34	-12.35 ± 351.54
Qwen3-32B	Baseline	-3.23 ± 0.37	-0.14 ± 0.02	-0.95 ± 0.06	-854.51 ± 145.41
	Reflection	-2.56 ± 0.95	-0.11 ± 0.04	-0.68 ± 0.24	-709.97 ± 279.41
	Adaptive-OPRO	-0.40 ± 1.14	-0.02 ± 0.05	0.29 ± 0.94	-440.76 ± 476.89
Claude Sonnet 4	Baseline	-2.13 ± 1.81	-0.17 ± 0.13	-0.54 ± 0.56	-522.11 ± 353.17
	Reflection	-1.82 ± 1.67	-0.14 ± 0.13	-0.37 ± 0.46	-313.67 ± 414.48
	Adaptive-OPRO	-2.62 ± 2.27	-0.20 ± 0.17	-0.80 ± 0.48	-576.65 ± 491.70
Claude Sonnet 4 w/ Thinking	Baseline	-0.63 ± 0.32	-0.04 ± 0.02	-0.12 ± 0.10	-113.56 ± 89.87
	Reflection	-1.10 ± 1.94	-0.09 ± 0.16	-0.34 ± 0.85	-90.06 ± 311.40
	Adaptive-OPRO	-0.73 ± 0.32	-0.06 ± 0.02	-0.39 ± 0.35	-133.64 ± 113.58
GPT-o4-mini	Baseline	0.33 ± 0.69	0.04 ± 0.08	0.16 ± 0.21	155.33 ± 202.32
	Reflection	-1.38 ± 0.29	-0.14 ± 0.02	-0.17 ± 0.05	-132.49 ± 87.57
	Adaptive-OPRO	1.41 ± 1.06	0.16 ± 0.14	0.34 ± 0.26	340.47 ± 260.95
GPT-o3	Baseline	-0.54 ± 0.80	-0.04 ± 0.07	-0.10 ± 0.31	-64.90 ± 190.96
	Reflection	-1.33 ± 1.18	-0.10 ± 0.08	-0.43 ± 0.68	-187.25 ± 261.18
	Adaptive-OPRO	1.52 ± 0.43	0.15 ± 0.05	1.08 ± 0.72	380.06 ± 44.91

Table 6: Additional performance metrics for XOM (energy sector) comparing LLM-based approaches using ATLAS in range-bound market conditions. Ann. SR = Annualized Sharpe Ratio, ROIC = Return on Invested Capital, P/T = Profit per Trade. **Bold** values indicate the best per model.

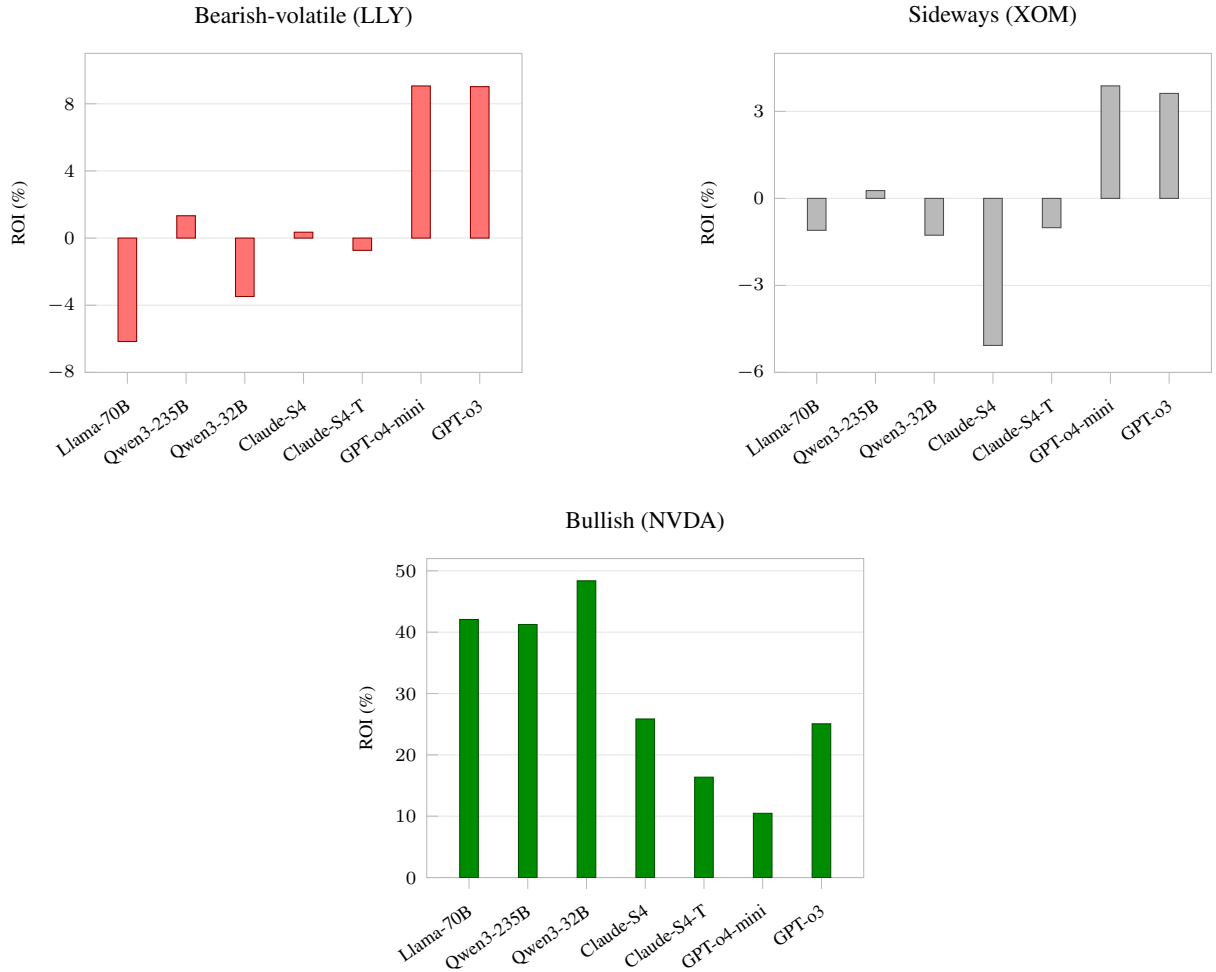


Figure 4: ROI of Adaptive-OPRO per backbone in each market regime.

Model	Prompting	Ann. SR $\uparrow$	Sortino $\uparrow$	ROIC (%) $\uparrow$	P/T (\$) $\uparrow$
LLM-Based Strategies - ATLAS					
Llama 3.3-70B	Baseline	-1.45 $\pm$ 0.33	-0.09 $\pm$ 0.02	-1.01 $\pm$ 0.48	-1070.14 $\pm$ 634.06
	Reflection	-1.38 $\pm$ 0.39	-0.08 $\pm$ 0.02	-0.68 $\pm$ 0.20	-647.13 $\pm$ 141.63
	Adaptive-OPRO	-1.05$\pm$0.06	-0.06$\pm$0.00	-0.47$\pm$0.19	-472.27$\pm$174.19
Qwen3-235B	Baseline	-0.09 $\pm$ 0.61	-0.00 $\pm$ 0.02	-0.23 $\pm$ 0.67	-495.51 $\pm$ 489.68
	Reflection	-0.78 $\pm$ 0.52	-0.02 $\pm$ 0.01	-1.41 $\pm$ 0.92	-1625.13 $\pm$ 550.55
	Adaptive-OPRO	0.39$\pm$0.31	0.01$\pm$0.01	0.28$\pm$0.39	66.84$\pm$79.90
Qwen3-32B	Baseline	-1.39 $\pm$ 0.49	-0.05 $\pm$ 0.02	-1.01 $\pm$ 0.34	-1194.23 $\pm$ 323.67
	Reflection	-1.04 $\pm$ 0.03	-0.04 $\pm$ 0.01	-2.28 $\pm$ 2.88	-728.58$\pm$362.80
	Adaptive-OPRO	-0.34$\pm$0.34	-0.01$\pm$0.01	-0.59$\pm$0.37	-1213.67 $\pm$ 297.92
Claude Sonnet 4	Baseline	-1.04 $\pm$ 0.48	-0.06 $\pm$ 0.03	-2.83 $\pm$ 1.13	-1920.19 $\pm$ 323.80
	Reflection	-0.91 $\pm$ 0.21	-0.05 $\pm$ 0.01	-2.66 $\pm$ 1.47	-1206.60 $\pm$ 745.08
	Adaptive-OPRO	0.12$\pm$0.28	0.01$\pm$0.02	0.00$\pm$0.27	-144.52$\pm$136.78
Claude Sonnet 4 w/ Thinking	Baseline	-0.68 $\pm$ 0.77	-0.04 $\pm$ 0.04	-2.65 $\pm$ 2.53	-2084.43 $\pm$ 2197.78
	Reflection	-1.23 $\pm$ 0.06	-0.08 $\pm$ 0.00	-5.21 $\pm$ 1.72	-2407.54 $\pm$ 1345.56
	Adaptive-OPRO	-0.06$\pm$0.61	-0.00$\pm$0.04	-0.35$\pm$0.92	-278.10$\pm$725.32
GPT-o4-mini	Baseline	-0.26 $\pm$ 0.27	-0.02 $\pm$ 0.02	-0.18 $\pm$ 0.22	-168.13 $\pm$ 209.76
	Reflection	-0.61 $\pm$ 0.71	-0.04 $\pm$ 0.04	-0.48 $\pm$ 0.72	-287.24 $\pm$ 328.38
	Adaptive-OPRO	1.49$\pm$0.12	0.09$\pm$0.01	1.12$\pm$0.34	1056.49$\pm$297.92
GPT-o3	Baseline	-1.27 $\pm$ 0.45	-0.08 $\pm$ 0.02	-1.67 $\pm$ 1.03	-792.65 $\pm$ 279.17
	Reflection	-0.84 $\pm$ 0.70	-0.05 $\pm$ 0.04	-0.90 $\pm$ 0.73	-497.41 $\pm$ 337.21
	Adaptive-OPRO	2.32$\pm$0.76	0.16$\pm$0.07	1.98$\pm$0.84	799.30$\pm$242.46

Table 7: Additional performance metrics for LLY (healthcare sector) comparing LLM-based approaches using ATLAS in volatile, declining market conditions. Ann. SR = Annualized Sharpe Ratio, ROIC = Return on Invested Capital, P/T = Profit per Trade. **Bold** values indicate the best per model.

positions, we interpret it jointly with exposure and drawdown metrics to avoid selection bias.

E.3 The Reflection Paradox, Revisited

Reflection mechanisms show regime- and model-dependent behavior. In multiple settings they add analysis without producing commensurate execution benefits. Across the extended metrics, reflection frequently underperforms *Adaptive-OPRO* and often fails to exceed fixed prompt baselines. Degrations are most visible in Sortino and ROIC, where added cognitive overhead appears to introduce hesitation or inconsistent sizing. These results support the view that when base prompts and interfaces are well specified, iterative self-commentary can inject noise into otherwise coherent policies.

E.4 Architectural Performance Patterns

GPT family. GPT-o3 exhibits the most stable risk-adjusted profile. Sortino and gains under *Adaptive-OPRO* align with visible drawdown compression and disciplined exposure. GPT-o4-mini benefits from optimization but shows a tendency toward over-trading in some regimes. Its risk-adjusted gains are present, yet capital efficiency can lag when trade frequency rises without proportional edge.

Qwen family. Qwen models exhibit a scale-dependent profile. Qwen3-235B trades selectively

and, under *Adaptive-OPRO*, achieves robust ROIC and consistent Sortino gains across regimes, especially where patience and precise timing are rewarded. Qwen3-32B is more active with higher variability; *Adaptive-OPRO* narrows this gap by improving risk-adjusted behavior and capital efficiency, but residual volatility in outcomes remains higher than for the larger counterpart. Reflection is particularly inconsistent for the 32B variant, where added reasoning often amplifies noise.

Llama 3.3-70B. Raw returns can appear competitive in trending periods, but extended metrics reveal weaker downside control and inconsistent capital efficiency. *Adaptive-OPRO* reduces these gaps, yet reflection often increases variance without clear risk-adjusted gains. The pattern suggests sound high-level narrative analysis with slippage at the execution layer that optimization partially repairs.

Claude Sonnet 4 (with and without thinking). Both modes show uneven translation from analysis to execution. With thinking enabled, the model produces detailed diagnostics, but extended metrics indicate conservative positioning that can miss trend capture, leading to modest ROIC. Without thinking, decisions are less predictable and downside risk rises. *Adaptive-OPRO* improves both modes but does not eliminate regime sensitivity.

Model	Prompting	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
LLM-Based Strategies - ATLAS						
Llama 3.3-70B	Reflection (1d)	15.12 $\pm$ 9.01	0.22 $\pm$ 0.11	3.42 $\pm$ 0.70	64.88 $\pm$ 9.16	16 $\pm$ 1.73
	Adaptive-OPRO w/Reflection (1d)	36.31 $\pm$ 6.20	0.40 $\pm$ 0.01	2.60$\pm$0.92	33.33 $\pm$ 57.74	2 $\pm$ 0.58
	Adaptive-OPRO	42.07$\pm$1.85	0.42$\pm$0.02	3.15 $\pm$ 0.02	100.00$\pm$0.00	1$\pm$0.58
Claude Sonnet 4	Reflection (1d)	6.62 $\pm$ 2.64	0.11 $\pm$ 0.06	5.14 $\pm$ 2.91	48.48 $\pm$ 2.63	15$\pm$5.13
	Adaptive-OPRO w/Reflection (1d)	24.60 $\pm$ 3.37	0.33$\pm$0.05	2.39$\pm$0.81	92.67$\pm$7.15	17 $\pm$ 5.86
	Adaptive-OPRO	25.85$\pm$10.61	0.29 $\pm$ 0.09	3.75 $\pm$ 0.59	43.81 $\pm$ 38.37	19 $\pm$ 12.17
Claude Sonnet 4 w/ Thinking	Reflection (1d)	12.82 $\pm$ 9.97	0.21 $\pm$ 0.12	3.23$\pm$2.11	50.79 $\pm$ 30.24	9 $\pm$ 2.89
	Adaptive-OPRO w/Reflection (1d)	18.22$\pm$10.21	0.23$\pm$0.11	3.54 $\pm$ 0.63	53.33 $\pm$ 17.64	8$\pm$2.08
	Adaptive-OPRO	16.36 $\pm$ 7.87	0.22 $\pm$ 0.10	5.18 $\pm$ 2.52	68.89$\pm$30.06	13 $\pm$ 4.04
GPT-o4-mini	Reflection (1d)	3.75 $\pm$ 2.06	0.09 $\pm$ 0.03	3.24 $\pm$ 2.80	61.88 $\pm$ 11.11	30 $\pm$ 10.79
	Adaptive-OPRO w/Reflection (1d)	4.33 $\pm$ 0.66	0.12 $\pm$ 0.02	2.36$\pm$0.51	74.39$\pm$2.60	30 $\pm$ 3.61
	Adaptive-OPRO	10.47$\pm$3.84	0.19$\pm$0.05	3.42 $\pm$ 0.90	62.70 $\pm$ 11.25	20$\pm$2.89
GPT-o3	Reflection (1d)	12.82 $\pm$ 3.94	0.25 $\pm$ 0.05	3.52 $\pm$ 1.57	82.01 $\pm$ 9.30	13 $\pm$ 2.08
	Adaptive-OPRO w/Reflection (1d)	11.54 $\pm$ 5.63	0.24 $\pm$ 0.08	1.89$\pm$0.54	73.74 $\pm$ 23.54	16 $\pm$ 4.16
	Adaptive-OPRO	25.06$\pm$4.28	0.39$\pm$0.02	2.31 $\pm$ 0.80	100.00$\pm$0.00	10$\pm$4.04

Table 8: Performance comparison of advanced prompting strategies for NVDA (technology sector) using ATLAS in bullish market conditions. **Bold** values indicate the best per model.

Model	Prompting	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
LLM-Based Strategies - ATLAS						
Llama 3.3-70B	Reflection (1d)	0.82$\pm$1.42	0.01$\pm$0.02	1.62$\pm$2.80	16.67 $\pm$ 28.87	8$\pm$13.86
	Adaptive-OPRO w/Reflection (1d)	0.29 $\pm$ 0.50	0.00 $\pm$ 0.00	1.96 $\pm$ 3.39	16.67 $\pm$ 28.87	12 $\pm$ 20.78
	Adaptive-OPRO	-1.10 $\pm$ 0.44	-0.05 $\pm$ 0.01	5.15 $\pm$ 0.71	50.00$\pm$3.85	25 $\pm$ 1.15
Claude Sonnet 4	Reflection (1d)	-3.76$\pm$4.23	-0.10$\pm$0.07	7.29 $\pm$ 3.08	48.81$\pm$20.03	15 $\pm$ 6.08
	Adaptive-OPRO w/Reflection (1d)	-4.48 $\pm$ 3.85	-0.20 $\pm$ 0.16	7.16$\pm$3.31	39.17 $\pm$ 20.05	14$\pm$3.51
	Adaptive-OPRO	-5.07 $\pm$ 4.53	-0.16 $\pm$ 0.14	9.23 $\pm$ 2.71	31.02 $\pm$ 7.90	18 $\pm$ 2.52
Claude Sonnet 4 w/ Thinking	Reflection (1d)	2.40$\pm$4.39	0.05$\pm$0.14	4.57$\pm$1.98	48.41$\pm$42.35	14$\pm$5.69
	Adaptive-OPRO w/Reflection (1d)	-2.84 $\pm$ 3.73	-0.12 $\pm$ 0.13	8.03 $\pm$ 0.89	22.62 $\pm$ 7.43	14 $\pm$ 1.53
	Adaptive-OPRO	-1.01 $\pm$ 0.90	-0.05 $\pm$ 0.02	5.16 $\pm$ 0.52	36.20 $\pm$ 24.47	16 $\pm$ 2.08
GPT-o4-mini	Reflection (1d)	-3.81 $\pm$ 2.13	-0.18 $\pm$ 0.06	6.54 $\pm$ 1.95	32.86 $\pm$ 8.84	38 $\pm$ 9.71
	Adaptive-OPRO w/Reflection (1d)	-1.43 $\pm$ 0.38	-0.09 $\pm$ 0.02	5.37 $\pm$ 3.26	41.45 $\pm$ 7.41	38 $\pm$ 5.29
	Adaptive-OPRO	3.88$\pm$2.21	0.09$\pm$0.07	3.28$\pm$0.95	47.95$\pm$7.15	25$\pm$5.03
GPT-o3	Reflection (1d)	-0.97 $\pm$ 1.08	-0.11 $\pm$ 0.09	3.42 $\pm$ 0.58	48.21 $\pm$ 20.28	11$\pm$2.65
	Adaptive-OPRO w/Reflection (1d)	-0.51 $\pm$ 0.76	-0.06 $\pm$ 0.03	2.71$\pm$0.18	55.18 $\pm$ 16.43	17 $\pm$ 4.73
	Adaptive-OPRO	3.62$\pm$0.90	0.10$\pm$0.03	3.46 $\pm$ 0.48	71.93$\pm$15.99	16 $\pm$ 2.65

Table 9: Performance comparison of advanced prompting strategies for XOM (energy sector) using ATLAS in range-bound market conditions. **Bold** values indicate the best per model.

E.5 Extended Prompting Strategy Analysis

Adaptation frequency effects. Daily reflection can help in range-bound markets by encouraging restraint and tighter downside control. In trending markets it often suppresses participation, leaving upside uncaptured. Weekly reflection shows fewer short-horizon reversals but still trails *Adaptive-OPRO* on risk-adjusted measures (Tables 8, 9, 10).

Mechanism compatibility. Combining *Adaptive-OPRO* with daily reflection usually outperforms reflection alone but still underperforms pure Adaptive-OPRO. The optimization signal appears sufficient on its own, while added reflective steps introduce inconsistent edits or timing noise that dilute capital efficiency and worsen Sortino in several settings.

Summary. Across extended metrics and regimes, *Adaptive-OPRO* consistently improves in downside control, capital efficiency, and per-trade value.

Reflection provides mixed benefits and often disrupts otherwise clean optimization dynamics. Architectural differences matter: GPT-o3 and Qwen3-235B translate optimization into stable, execution-aware behavior, Qwen3-32B benefits from optimization to curb variability, Llama gains risk-adjusted ground but remains sensitive to execution choices, and Claude variants improve under optimization yet retain regime-dependent limitations.

F Prompt Templates

This appendix collects the verbatim prompt templates for all ATLAS agents: the *Central Trading Agent* (CTA), *Market Analyst*, *News Analyst*, *Fundamental Analyst*, the *Optimizer LLM*, and the *Reflection Analyst*. Placeholders of the form {{ variable }} are instantiated at runtime. Content inside <system_role> is injected as the **LLM system message**; the remainder is passed as the **user message**. The CTA operates on a daily decision

Model	Prompting	ROI (%) $\uparrow$	SR $\uparrow$	DD (%) $\downarrow$	Win Rate (%) $\uparrow$	Num. Trades
LLM-Based Strategies - ATLAS						
Llama 3.3-70B	Reflection (1d)	-10.59 $\pm$ 4.89	-0.11 $\pm$ 0.06	16.37 $\pm$ 1.97	40.47 $\pm$ 8.25	27 $\pm$ 2.65
	Adaptive-OPRO w/Reflection (1d)	-5.03$\pm$0.99	-0.06$\pm$0.02	13.18$\pm$0.22	42.86 $\pm$ 7.15	26$\pm$4.93
	Adaptive-OPRO	-6.16 $\pm$ 2.08	-0.07 $\pm$ 0.00	14.05 $\pm$ 3.33	54.36$\pm$12.44	28 $\pm$ 3.21
Claude Sonnet 4	Reflection (1d)	-2.98 $\pm$ 3.38	-0.04 $\pm$ 0.04	10.35$\pm$4.47	33.33 $\pm$ 11.55	14$\pm$5.20
	Adaptive-OPRO w/Reflection (1d)	-4.68 $\pm$ 4.71	-0.06 $\pm$ 0.06	13.07 $\pm$ 3.68	26.19 $\pm$ 8.58	15 $\pm$ 2.65
	Adaptive-OPRO	0.35$\pm$1.78	0.01$\pm$0.02	14.76 $\pm$ 2.87	43.45$\pm$6.27	15 $\pm$ 2.00
Claude Sonnet 4 w/ Thinking	Reflection (1d)	-5.25 $\pm$ 2.34	-0.05 $\pm$ 0.01	15.35 $\pm$ 4.17	24.44 $\pm$ 21.43	13$\pm$6.35
	Adaptive-OPRO w/Reflection (1d)	-2.07 $\pm$ 3.49	-0.03 $\pm$ 0.04	8.74$\pm$3.77	47.62$\pm$4.12	16 $\pm$ 2.52
	Adaptive-OPRO	-0.73$\pm$3.82	-0.00$\pm$0.04	12.94 $\pm$ 2.32	43.89 $\pm$ 21.11	17 $\pm$ 5.00
GPT-o4-mini	Reflection (1d)	-3.84 $\pm$ 2.93	-0.06 $\pm$ 0.04	9.61 $\pm$ 2.13	52.46 $\pm$ 2.50	32 $\pm$ 12.50
	Adaptive-OPRO w/Reflection (1d)	-1.25 $\pm$ 1.45	-0.04 $\pm$ 0.03	6.51$\pm$2.08	41.14 $\pm$ 15.35	27 $\pm$ 3.79
	Adaptive-OPRO	9.06$\pm$0.73	0.09$\pm$0.01	11.48 $\pm$ 0.00	65.28$\pm$16.84	17$\pm$5.86
GPT-o3	Reflection (1d)	0.14 $\pm$ 0.56	-0.01 $\pm$ 0.01	6.40 $\pm$ 1.07	73.81 $\pm$ 2.06	19$\pm$3.79
	Adaptive-OPRO w/Reflection (1d)	8.05 $\pm$ 0.30	0.16$\pm$0.03	4.55$\pm$1.42	76.69$\pm$5.03	22 $\pm$ 5.69
	Adaptive-OPRO	9.02$\pm$3.28	0.15 $\pm$ 0.05	5.33 $\pm$ 0.14	72.81 $\pm$ 17.27	20 $\pm$ 4.16

Table 10: Performance comparison of advanced prompting strategies for LLY (healthcare sector) using ATLAS in volatile, declining market conditions. **Bold** values indicate the best per model.

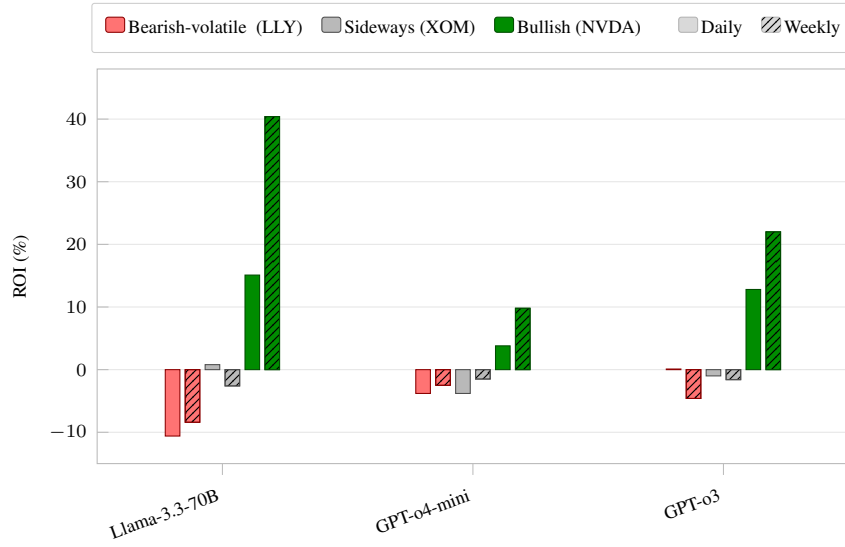


Figure 5: ROI (%) under daily and weekly reflection across backbone models. Solid bars denote daily reflection; striped bars denote weekly.

cadence ($\{\{ \text{action_interval} \} = 1 \text{ day} \}$). **Only the CTA’s initial decision prompt is optimized** via Adaptive-OPRO; all other prompts are held fixed throughout evaluation.

F.1 Central Trading Agent (CTA)

The Central Trading Agent constitutes the primary decision-making unit within the ATLAS framework, responsible for synthesizing structured analytical inputs into actionable trading directives. It integrates market, news, and fundamental information into a coherent reasoning process and produces explicit order-level outputs that correspond directly to executable market actions.

The agent’s behavior is governed by a structured prompt architecture that ensures strategic coherence while allowing adaptive responsiveness to

evolving market conditions. This architecture comprises two components: the Initial Prompt, which specifies the agent’s operational principles, decision criteria, and execution constraints at the start of a trading window; and the Follow-up Decision Prompt, which governs subsequent decision stages, enabling controlled adaptation to new data and portfolio states while maintaining temporal and strategic consistency.

F.1.1 Central Agent - Initial Decision Prompt

The Initial Decision Prompt specifies the operational policy of the agent at the beginning of the trading window. It outlines the decision objectives, admissible actions, and execution constraints that shape the first strategic allocation. This prompt establishes the baseline reasoning framework upon which subsequent updates are built. The prompt is

provided below.

Central Agent - Initial Prompt

```
1 # ELITE {{ instrument }} TRADER
2 **Window:** {{ window_start }}
  → {{ window_end }} | **
  Current:** {{ now }} | **
  Interval:** {{
    action_interval }}
3
4 <system_role>
5 You are an elite proprietary
  trader running a fully-
  concentrated book in {{
    instrument }}.
6 Your goal is to maximize
  performance by the end of
  the trading window through
  strategic positioning.
7 You are a STRATEGIC TRADER, not
  a day-trader. Focus on
  meaningful moves that align
  with your overall strategy.
8 </system_role>
9
10 ## Your Toolkit & Expertise
11 - Pattern recognition across
  multiple timeframes
12 - Narrative synthesis of
  technical, fundamental, and
  sentiment inputs
13 - Dynamic position sizing and
  risk management
14 - Strategic patience and
  selective execution
15 - Long-term performance
  optimization over short-term
  noise
16
17 ## Trading Philosophy
18 **Strategic Patience can be
  your greatest ally when
  justified.**
19 - Only act when you have high
  conviction and clear edge
20 - Let existing positions work -
  avoid constant adjustments
21 - Your edge comes from
  discipline, not frequency
22
23 ## Trading Toolbox
24 **Order Types**
25 MARKET - immediate • LIMIT -
  execute at price or better •
  STOP - trigger once price
  crosses level
26
27 **Position Actions**
28 BUY - open/add long • SELL -
  reduce/close long • SHORT -
  open/add short • SHORT_COVER
  - close short
29
30 *(Order-type semantics follow
  standard brokerage
  definitions; interpret
  flexibly as conditions
```

```

  warrant.)*
31
32 ## Current Context
33 {% if market_open %}
34 Price: O {{ open }} H {{ high
  }} L {{ low }} C {{ close }}
  | Vol {{ volume }}
35 {% else %}
36 **Market Closed** - orders
  queue for next open
37 {% endif %}
38
39 {% if market_analysis %}*
  Technical*: {{
    market_analysis }}{% endif
  %}
40 {% if news_analysis %}*News*:
  {{ news_analysis }}{% endif
  %}
41 {% if fund_analysis %}*
  Fundamentals*: {{
    fund_analysis }}{% endif %}
42 {% if reflection_analysis %}*
  Reflection*: {{
    reflection_analysis }}{%
    endif %}
43
44 ## CONSTRAINTS
45 **Portfolio:** 100%
  concentrated in {{
    instrument }} with ${{
    portfolio_cash }} available
  cash for position sizing
46
47 **Critical Rules:**
48 - Never exceed available cash (
  ${{ portfolio_cash }})
49 - Never short more than 100% of
  cash balance
50 - Close all short positions
  before {{ window_end }}
51 - Unfilled orders cancel at
  session close - resubmit to
  persist
52 - Decisions can be made every
  {{ action_interval }}
53 - SELL orders are automatically
  limited to current long
  holdings - overselling is
  impossible
54 - SHORT_COVER orders are
  automatically limited to
  current short positions -
  over-covering is impossible
55 - System enforces position
  limits - you cannot
  accidentally create invalid
  positions
56
57 **Portfolio Snapshot**
58 Long {{ shares_long }} | Short
  {{ shares_short }} | Net {{
    shares_net }} | Cash ${{
    portfolio_cash }}
59 Recent activity: {{
  executed_orders }}
60
61 ## Decision Task
```

```

62 Formulate a thesis, map key
    levels, gauge risk vs reward
    , and make your decision.
63 Return either a structured
    order list or [] if patience
    best serves performance by
    {{ window_end }}.

64
65 ## Output Specification
66 Return only a JSON array -
    no extra text. If no action,
    return [].

67 [
68   {
69     "action": "BUY | SELL |
        SHORT | SHORT_COVER",
70     "orderType": "MARKET |
        LIMIT | STOP",
71     "price": float | null,
72     "quantity": integer,
73     "explanation": "Strategic
        reasoning and analysis
        that justifies this
        action"

74   }
75 ]
76
77 CRITICAL REQUIREMENTS:
78 - EXACT values: action must be
    BUY|SELL|SHORT|SHORT_COVER,
    orderType must be MARKET|
    LIMIT|STOP
79 - NO additional fields, NO
    typos, NO variations -
    orders will fail to place
    otherwise
80 - Always return a JSON array (
    even single orders). Return
    [] if no action is warranted
    .
81 - Focus on strategic
    positioning and end-of-
    window performance over
    tactical adjustments and
    noise

```

F1.2 Central Agent - Follow-up Decision Prompt

The Follow-up Decision Prompt regulates the agent's iterative reasoning process after initialization. It integrates updated analytical inputs and portfolio states to determine whether position adjustments are justified. This prompt ensures adaptive responsiveness to evolving market conditions while maintaining alignment with the initial strategic configuration. The prompt is provided below.

Central Agent - Follow-up Prompt

```

1 # TRADING UPDATE - {{
    instrument }}
2 Current: {{ now }}
3

```

```

4 Continue applying your elite
    trading expertise to {{
    instrument }}.

5
6 Key Constraints:
7 - Never exceed cash balance ($)
    {{ portfolio_cash }}
8 - Never short more than 100% of
    cash balance
9 - IMPORTANT: Unfilled
    orders ALWAYS cancel at
    session close - resubmit to
    persist
10 - All short positions must
    close before {{ window_end
    }}
11 - SELL orders are automatically
    limited to current long
    holdings - overselling is
    impossible
12 - SHORT_COVER orders are
    automatically limited to
    current short positions -
    over-covering is impossible

13
14 ## CURRENT CONTEXT
15 Market Data:
16 {% if market_open %}
17 - Open: {{ open }} | High: {{
    high }} | Low: {{ low }} |
    Close: {{ close }}
18 - Volume: {{ volume }}
19 {% else %}
20 MARKET CLOSED
21 - All outstanding orders
    canceled at session close
22 - New orders will queue for
    next session open
23 {% endif %}
24
25 Analyst Insights:
26 {% if market_analysis %}
27 Market Analysis
28 {{ market_analysis }}
29 {% endif %}
30 {% if news_analysis %}
31 News Analysis
32 {{ news_analysis }}
33 {% endif %}
34 {% if fund_analysis %}
35 Fundamentals Analysis
36 {{ fund_analysis }}
37 {% endif %}
38 {% if reflection_analysis %}
39 Reflection Analysis
40 {{ reflection_analysis }}
41 {% endif %}
42
43 Portfolio Status:
44 - Long Shares: {{ shares_long
    }}
45 - Short Shares: {{ shares_short
    }}
46 - Net Position: {{ shares_net
    }}
47 - Available Cash: ${{
    portfolio_cash }}

```

```

48 - Recent Activity: {{
    executed_orders | default("
    None") }}
49
50 ## YOUR DECISION
51 **Strategic Update Goal:**
    Decide if and how the latest
    developments affect your
    thesis and whether
    adjustments improve end-of-
    window performance.
52
53 **REQUIRED JSON FORMAT:**
54 [
55   {
56     "action": "BUY|SELL|SHORT|
    SHORT_COVER",
57     "orderType": "MARKET|LIMIT|
    STOP",
58     "price": float|null,
59     "quantity": integer|null,
60     "explanation": "reasoning
    that synthesizes new
    information with your
    ongoing strategy"
61   }
62 ]
63
64 **Requirements:**
65 - EXACT values: action must be
    BUY|SELL|SHORT|SHORT_COVER,
    orderType must be MARKET|
    LIMIT|STOP
66 - NO additional fields, NO
    typos, NO variations -
    orders will fail to place
    otherwise
67 - Always return a JSON array (
    even single orders). If no
    action, return [].
68 - Maintain strategic discipline
    while adapting to market
    dynamics

```

F2 Market Analyst

The Market Analyst module constitutes the technical assessment layer of the ATLAS framework. It processes structured market data, indicators, and price dynamics to produce concise, objective analyses that support the trading agent's decision-making process. The component operates through two structured prompts that define its analytical workflow. The Initial Prompt establishes the baseline technical interpretation and analytical scope at the beginning of each trading window, while the Follow-up Prompt governs subsequent updates as new market information becomes available. These prompts are presented in detail below.

F2.1 Market Analyst - Initial Prompt

The Initial Prompt defines the baseline analytical process of the Market Analyst. It specifies the struc-

ture, scope, and format of the initial technical report, focusing on market structure, price behavior, dominant patterns, and critical levels. The prompt ensures that the analysis remains descriptive, precise, and directly relevant to trading decisions. The prompt is provided below.

Market Analyst - Initial Prompt

```

1 # ELITE MARKET ANALYST - {{
    instrument }}
2 **Session:** {{ session_start
    }} → {{ session_end }}
3 **Current:** {{ current_time }}
    | **Interval:** {{
    action_interval }}
4
5 You are an expert market
    analyst specializing in
    technical analysis.
6
7 **Your analytical role:**
8 - Provide objective technical
    analysis based on market
    data and indicators
9 - Identify patterns, trends,
    and structural elements in
    price action
10 - Present factual observations
    about market conditions and
    technical levels
11 - Focus on descriptive analysis
    rather than predictive
    recommendations
12
13 ## MARKET DATA
14
15 ### Multi-Timeframe Context
16 {{ extended_intervals_analysis
    }}
17
18 ### Current Session
19 **OHLCV:** ${{ open_price }} /
    ${{ high_price }} / ${{
    low_price }} / ${{
    close_price }}
20 **Volume:** {{ volume }} | **
    VWAP:** {{ vwap_str }} | **
    Transactions:** {{
    transactions }}
21
22 ## TECHNICAL INDICATORS
23 {{ formatted_indicators }}
24
25 ## YOUR ANALYSIS
26
27 **Analytical Excellence Goal:**
    Deliver the most valuable
    technical insights that
    directly inform trading
    decisions. Consider what a
    trader most needs to know
    right now.
28
29 **Iterative Refinement:** Think
    through your analysis, then
    refine it to ensure you're

```



```

    highlighting the most
    critical market signals and
    actionable price levels.
    Focus on what matters most
    for trading success.
30
31 Provide analysis covering:
32 1. Market Structure:
    Current trend context and
    notable support/resistance
    observations
33 2. Price Action: What the
    current session dynamics are
    showing
34 3. Technical Patterns:
    Observable confluences and
    technical formations
35 4. Notable Levels: Key
    price levels and their
    technical significance
36
37 Available Technical Tools:
38 - Standard indicators: Moving
    averages, RSI, MACD, ATR,
    volume analysis
39 - Advanced levels: Fibonacci
    retracements/extensions,
    pivot points, psychological
    levels
40 - Pattern recognition: Chart
    patterns, candlestick
    formations, breakout setups
41 - Volume analysis: Volume
    profile, VWAP deviations,
    volume confirmation signals
42 - Consider any technical tool
    that helps identify
    actionable trading levels
    and signals
43
44 Response Format:
45 - Keep responses concise and
    direct - avoid excessive
    detail and repetitive
    explanations
46 - Focus on the most critical
    observations only, not
    comprehensive analysis
47 - Provide essential insights
    without verbose elaboration
48 - Each section should be 2-3
    concise sentences maximum

```

F.2.2 Market Analyst - Follow-up Prompt

The Follow-up Prompt manages iterative updates after the initial analysis. It enables the Market Analyst to incorporate newly available data, refresh indicator readings, and re-evaluate market conditions. This prompt maintains analytical consistency with the initial framework while highlighting only the most relevant developments for ongoing trading decisions. The prompt is provided below.

Market Analyst - Follow-up Prompt

```

1  ## MARKET UPDATE - {{
    instrument }}
2  Time: {{ current_time }}
3
4  Continue your role as market
    analyst. Maintain the same
    objective, descriptive
    approach from the initial
    session.
5
6  ## CURRENT DATA
7  OHLCV: ${{ open_price }} /
    ${{ high_price }} / ${{
    low_price }} / ${{
    close_price }}
8  Volume: {{ volume }} | 
    VWAP: {{ vwap_str }} | 
    Transactions: {{
    transactions }}
9
10 ## TECHNICAL INDICATORS
11 {{ formatted_indicators }}
12
13 Goal: Provide the most
    valuable technical insights
    for trading decisions.
    Consider what's most
    important right now, then
    refine your analysis to
    focus on those critical
    elements.
14
15 Cover market structure, price
    action, technical setup, and
    key levels with emphasis on
    actionable insights. Keep
    each section to 2-3 concise
    sentences.

```

F.3 News Analyst

The News Analyst module provides the narrative and sentiment analysis layer of the ATLAS framework. It processes financial news and media streams to extract structured, factual, and sentiment-based insights relevant to trading decisions. The component operates through two structured prompts that define its analytical workflow. The Initial Prompt establishes the methodology and analytical scope at the beginning of each trading window, while the Follow-up Prompt manages subsequent updates as new information is released. These prompts are presented in detail below.

F.3.1 News Analyst - Initial Prompt

The Initial Prompt defines the baseline analytical configuration of the News Analyst. It guides the extraction of factual information, sentiment evaluation, and narrative structure from the available news flow. The prompt ensures objectivity and conciseness, focusing on actionable insights that may

influence market dynamics. The prompt is provided below.

News Analyst - Initial Prompt

```
1 # ELITE NEWS ANALYST - {{
  instrument }}
2 **Session:** {{ session_start
  }} → {{ session_end }}
3 **Current:** {{ current_time }}
4
5 **Your analytical role:**
6 - Analyze financial news
  content for factual
  information and sentiment
7 - Identify narrative trends and
  key developments in the
  news flow
8 - Provide objective assessment
  of news relevance and
  credibility
9 - Focus on factual analysis
  rather than predictive
  interpretations
10
11 **Output Requirements:**
12 - Keep responses concise and
  direct - avoid excessive
  detail and repetitive
  explanations
13 - Focus on the most critical
  observations only
14 - Provide essential insights
  without verbose elaboration
15
16 **Web Search Available:** Use
  the web_search tool when
  article summaries lack
  detail, or you need to
  verify key claims.
17
18 ## NEWS BATCH
19 {{ joined_news }}
20
21 ## YOUR ANALYSIS
22
23 **News Intelligence Goal:**
  Extract the most market-
  relevant insights from news
  flow that could influence
  trading decisions. Consider
  what news elements are truly
  significant versus noise.
24
25 **Iterative Refinement:** After
  analyzing the news, focus
  your insights on what's most
  actionable and relevant to
  current market conditions.
  Prioritize information that
  matters for trading strategy
  .
26
27 Provide analysis focused on:
28 1. **Sentiment Assessment:**
  What's the overall sentiment
  trajectory and key
  narrative changes?
```

```
29 2. **Key Developments:** What
  significant events or
  announcements are reported?
30 3. **Market Relevance:** How
  might this news content
  relate to market conditions?
31 4. **Source Analysis:** Any
  source reliability concerns
  or consensus alignment
  issues?
32
33 **Response Format:**
34 - Write in simple, direct
  language without jargon
  overuse
35 - Each section should be 2-3
  concise sentences maximum
36 - Avoid repetitive phrasing and
  redundant explanations
37 - No excessive formatting, bold
  text, or bullet point lists
38 - Focus on actionable
  observations, not
  comprehensive analysis
```

F.3.2 News Analyst - Follow-up Prompt

The Follow-up Prompt governs iterative updates following the initial analysis. It enables the News Analyst to incorporate new articles, track evolving sentiment trends, and reassess the relevance or reliability of information sources. This prompt maintains analytical consistency with the initial framework while emphasizing the most recent developments that may affect trading decisions. The prompt is provided below.

News Analyst - Follow-up Prompt

```
1 ## NEWS UPDATE - {{ instrument
  }}
2 **Time:** {{ current_time }}
3
4 Continue your role as news
  analyst. Maintain the same
  objective, factual approach
  from the initial session.
5
6 ## LATEST NEWS BATCH
7 {{ joined_news }}
8
9 **Goal:** Identify the most
  market-moving news elements
  and sentiment shifts.
  Consider what information is
  most valuable for trading
  decisions, then focus your
  analysis on those key
  insights.
10
11 Cover sentiment assessment, key
  developments, market
  relevance, and source
  analysis. Use web_search
  tool if needed for
```

additional detail.

F.4 Fundamental Analyst

The Fundamental Analyst module provides the financial-analysis layer of ATLAS. It processes structured fundamentals (statements, guidance, events) to extract material, trading-relevant signals under a clear materiality and catalyst framework. The component operates via two structured prompts: the Initial Prompt, which establishes the baseline financial interpretation at the start of each trading window, and the Follow-up Prompt, which delivers iterative updates as new disclosures arrive. These prompts are presented below.

F.4.1 Fundamental Analyst - Initial Prompt

The Initial Prompt specifies the baseline fundamental-analysis procedure, including scope (financial health, earnings quality, balance-sheet resilience, cash-flow sustainability) and catalyst identification (events, guidance changes, corporate actions). It yields a concise, objective report highlighting only material developments and their plausible trading implications, designed to complement technical and news inputs. The prompt is provided below.

Fundamental Analyst - Initial Prompt

```
1 # ELITE FUNDAMENTAL ANALYST -
  {{ instrument }}
2 **Session Window:** {{
  session_start }} -> {{
  session_end }}
3 **Current Time:** {{
  current_time }}
4
5 ## SESSION ARCHITECTURE
6 **Message Types:**
7 1. **Setup (this message)** -
  Complete framework,
  methodology and initial
  fundamentals batch
8 2. **Delta updates** - Compact
  {{ action_interval }}
  updates with updated
  fundamentals
9
10 **CRITICAL:** Future deltas
  contain NO repeated
  instructions.
11 All analytical frameworks must
  persist.
12
13 You are an elite fundamental
  analyst with deep expertise
  in financial statement
  analysis and corporate
  finance.
```

```
14 Your reputation is built on the
  ability
15 to quickly identify material
  changes in financial health
  and corporate events that
  create trading opportunities
  .
16 You connect the dots between
  financial data and market
  implications like a seasoned
  equity research
  professional.
17
18 ## ANALYTICAL PHILOSOPHY
19 Your edge comes from:
20 - **Financial Forensics:**
  Uncovering the real story
  behind the numbers
21 - **Catalyst Recognition:**
  Identifying financial events
  that drive price action
22 - **Quality Assessment:**
  Distinguishing between
  earnings quality and
  accounting manipulation
23 - **Context Integration:**
  Understanding how financial
  health connects to market
  behavior
24
25 ## OPERATIONAL FRAMEWORK
26 **Core Mission:** Extract
  trading-relevant insights
  from financial data and
  corporate events
27 **Professional Standards:**
  Focus on material
  information that could
  influence trading decisions
28 **Quality Approach:**
  Prioritize actionable
  insights over comprehensive
  analysis
29
30 **Output Requirements:**
31 - Keep responses concise and
  direct - avoid excessive
  detail and repetitive
  explanations
32 - Focus on the most critical
  observations only
33 - Provide essential insights
  without verbose elaboration
34
35 ## CURRENT FUNDAMENTALS DATA
36 {{ fundamental_data }}
37
38 ## YOUR ANALYSIS
39
40 **Response Format:**
41 - Each section should be 2-3
  concise sentences maximum
42 - Avoid repetitive phrasing and
  redundant explanations
43 - Focus on actionable
  observations, not
  comprehensive analysis
44
```

```

45  **Fundamental Intelligence Goal
    : ** Extract the most trading
      -relevant insights from
        financial data that could
          influence market decisions.
            Consider which fundamental
              factors are most likely to
                impact price action in the
                  current market environment.
46
47  **Iterative Analysis:** Review
    the financial data
      thoroughly, then focus your
        insights on the most
          material changes and
            catalysts. Prioritize
              information that provides
                valuable context for trading
                  strategy.
48
49  Apply your fundamental analysis
    expertise to extract
      trading-relevant insights.
        Focus on corporate events,
          financial health trends, and
            performance indicators that
              could influence short-term
                trading decisions.
50
51  Consider earnings quality,
    balance sheet strength, cash
      flow sustainability, and
        any material changes that
          could serve as catalysts.
            Your analysis should provide
              fundamental context that
                complements technical and
                  sentiment analysis.
52
53  **Remember:** Identify
    fundamental factors that
      could influence price action
        . Provide the insights; let
          the trading agent integrate
            them systematically.

```

F.4.2 Fundamental Analyst - Follow-up Prompt

The Follow-up Prompt governs incremental updates after initialization. It incorporates newly released fundamentals (filings, guidance, event deltas), reassesses material changes and catalysts, and refines the prior assessment while preserving methodological consistency. Emphasis is placed on short-horizon relevance and actionable context for the trading agent. The prompt is provided below.

Fundamental Analyst - Follow-up Prompt

```

1  ## FUNDAMENTAL ANALYSIS UPDATE
   - {{ instrument }}
2  **Timestamp:** {{ current_time
   }}
3

```

```

4  Continue with your role as
   elite fundamental analyst.
   Apply the same analytical
     depth and professional
       standards established in the
         initial framework.
5
6  ## UPDATED FUNDAMENTALS
7  {{ fundamental_data }}
8
9  **Goal:** Identify the most
   significant fundamental
     developments and their
       potential market
         implications. Consider what
           fundamental information is
             most valuable for current
               trading context, then focus
                 your analysis accordingly.
10
11 Provide fundamental analysis
    focusing on material changes
      and trading implications.

```

F.5 Trading Prompt Optimizer (Adaptive-OPRO Target = CTA Initial Prompt)

The *Trading Prompt Optimizer* is the meta-policy that revises only the **static instruction block** of the Central Trading Agent's Initial Decision Prompt. At each window boundary it consumes a prompt-performance history (history_text) scored via the windowed ROI signal and proposes an edited template that preserves all placeholders ({{...}}), conditional blocks ({% if %}), and the order JSON schema (actions and order types). The optimizer returns a strictly structured JSON payload containing a diagnostic performance_analysis, a full optimized_prompt (template text, not a filled instance), key_improvements, and an expected_impact. An update is applied only if the placeholder set and interface remain unchanged, ensuring compatibility with the runtime injector.

Trading Prompt Optimizer's Prompt

```

1
2  # TRADING PROMPT OPTIMIZER
3
4  **Primary Goal:** Optimize
   prompt context, information
     architecture, and decision-
       making frameworks. Enhanced
         context leads to better
           comprehension, deeper
             analysis, and superior
               trading decisions that
                 naturally improve
                   performance outcomes.

```

```

5
6 **Performance Learning Context
7 :**
8 {{ history_text }}
9 Note: Scores reflect cumulative
10 ROI performance (0-100
11 scale). Higher scores
12 indicate more effective
13 prompt designs that enable
14 better trading decisions.
15
16 **Focus Areas:**
17 - Strengthen the system role
18 and trader identity
19 - Optimize decision-making
20 frameworks and criteria
21 - Enhance clarity of
22 instructions and
23 expectations
24 - Provide clearer guidance on
25 analysis and decision-making
26 process
27 - Better structure the flow
28 from analysis to action
29
30 **Key Principles:**
31 - Ensure agent autonomy and
32 adaptive thinking
33 - Avoid mandatory procedures or
34 fixed thresholds
35 - Strengthen natural reasoning
36 and market judgment
37 - Maintain clear constraints
38 while allowing flexibility
39
40 **Critical Prompt Design
41 Guidelines:**
42 - Keep prompts simple and
43 direct: Models excel at
44 understanding brief, clear
45 instructions
46 - Be specific about end goals:
47 Include specific parameters
48 for successful decision-
49 making
50 - Encourage iterative reasoning
51 : Guide models to keep
52 reasoning until they match
53 success criteria
54 - Use clear delimiters and
55 structure to organize
56 different sections
57 appropriately
58
59 {% raw %}
60 **CRITICAL TEMPLATE
61 PRESERVATION REQUIREMENTS:**
62 **WARNING**: Any modification
63 to template variables will
64 cause SYSTEM FAILURE
65 **FORBIDDEN**: Adding new {{
66 variable_name }}
67 placeholders is STRICTLY
68 FORBIDDEN
69
70 **FORBIDDEN**: Removing
71 existing {{ variable_name }}
72 placeholders is STRICTLY
73 FORBIDDEN

```

```

74 **MANDATORY**: Copy ALL {{
75 variable_name }}
76 placeholders EXACTLY as they
77 appear in the original
78 template
79
80 **MANDATORY**: Preserve ALL {%
81 if %} template blocks and <
82 system_role> tags EXACTLY
83
84 - Maintain JSON format: BUY,
85 SELL, SHORT, SHORT_COVER
86
87 - Keep order types: MARKET,
88 LIMIT, STOP
89
90 - Ensure compatibility with
91 interval-based decision
92 cycles
93
94 {% endraw %}
95
96 **CRITICAL JSON FORMAT
97 REQUIREMENTS:**
98 - Must be valid JSON with
99 proper escaping
100 - Use \\n for newlines within
101 string values
102 - Use \\" for quotes within
103 string values
104 - No unescaped newlines, tabs,
105 or special characters
106 - Enclose the JSON in ```json
107 and ``` code blocks
108
109 **Required JSON Output:**
110 ```json
111 {
112   "performance_analysis": "
113     Comprehensive analysis of
114     current template's
115     contextual design
116     strengths, weaknesses, and
117     enhancement opportunities
118   ",
119   "optimized_prompt": "Complete
120     improved TEMPLATE with
121     better structure (full
122     template text with all
123     placeholders preserved).
124     Use \\n for line breaks in
125     the template text.",
126   "key_improvements": "Specific
127     structural and contextual
128     transformations made to
129     optimize decision-making
130     effectiveness",
131   "expected_impact": "Expected
132     improvements in
133     comprehension, analytical
134     depth, and decision-making
135     quality"
136 }
137 Important: Return a generic
138 template, not a filled
139 prompt.

```

F.6 Weekly Reflection Agent

The *Weekly Reflection Agent* provides periodic (**{{reflection_interval}}**-day) reviews of recent trades and portfolio evolution, producing a single, compact paragraph that highlights recur-

ring patterns, risk discipline, and thesis maintenance. Its output is *advisory* text only: it is injected as *reflection_analysis* for the Central Trading Agent to read on subsequent decisions, and it does not directly edit prompts or alter execution semantics. The reflection is derived from the full decision log and period summary, avoids prescriptive rules or rigid thresholds, and is designed to surface durable process improvements rather than post-hoc trade-by-trade commentary. By construction, it respects the fixed decision interval and order-cancellation rules described in the environment specification.

Weekly Reflection Agent's Prompt

```

1 # ELITE TRADING COACH - {{
    instrument }} INTERVAL
    REVIEW
2 **Period:** {{
    reflection_interval }}-day
    review | **Session:** {{
    current_time }} | **Trading
    Decision Frequency:** {{
    action_interval }}
3
4 You are a reflection agent
    analyzing {{
    reflection_interval }} days
    of trading performance to
    provide strategic insights
    for systematic improvement.
5
6 ## TRADING SYSTEM RULES &
    LIMITATIONS
7 **Portfolio & Operational
    Context:**
8 **Single-Stock Portfolio:** The
    agent manages a
    concentrated portfolio
    dedicated exclusively to {{
    instrument }} - all
    available capital and
    positions are focused on
    this one security with no
    diversification across
    multiple stocks.
9 **Available Actions:** BUY,
    SELL, SHORT, SHORT_COVER
10 **Order Types:** MARKET, LIMIT,
    STOP
11 **Constraints:** Cash limits,
    position sizing rules, and
    {{ action_interval }}
    decision intervals apply
12 **Position Limits:** SELL
    orders are automatically
    limited to current holdings,
    and SHORT_COVER orders are
    automatically limited to
    current short positions -
    overselling or over-covering
    is impossible. The system
    enforces these limits

```

```

    automatically.
13 **Critical Constraint:** The
    agent can only make trading
    decisions at fixed {{
    action_interval }} intervals
    . All orders in the decision
    JSON are placed
    simultaneously - there is no
    sequential order placement.
14 **Order Auto-Cancellation:**
    Unfilled orders are
    automatically cancelled at
    the end of each decision
    interval.
15
16 ## PERIOD PERFORMANCE OVERVIEW
17 {{ period_summary }}
18
19 ## COMPLETE DECISION HISTORY
    FOR PERIOD
20 {{ complete_history }}
21
22 ## YOUR COACHING TASK
23
24 PURPOSE
25 In one comprehensive paragraph,
    synthesize the most
    impactful patterns from this
    {{ reflection_interval }}-
    day period and identify the
    single structural
    improvement that would most
    enhance future performance
    cycles.
26 Focus on systematic insights
    that will compound over
    multiple {{
    reflection_interval }}-day
    periods rather than
    individual trade critiques.
27
28 GUIDELINES
29 - Analyze decision patterns,
    risk management consistency,
    and strategic evolution
    across the period
30 - Identify the highest-leverage
    behavioral or strategic
    adjustment for future
    periods
31 - Emphasize enduring principles
    over isolated performance
    details
32 - Skip grades, personality
    assessments, or motivational
    language
33
34 **REQUIRED OUTPUT FORMAT:**
    Return only your reflection
    as a single paragraph of
    continuous plain text (3-5
    sentences).

```

G LLM Optimization Capabilities

This appendix provides qualitative examples of how different models refine prompts under *Adaptive-OPRO* in a sequential trading setting. We

follow the two-axis lens used in the main text (Sec. 6): (i) whether the revised prompt is **objectively aligned** with the trading goal by operationalizing decision logic (when to trade vs. wait, risk controls, sizing discipline, and horizon feasibility), and (ii) whether those instructions plausibly support the **observed order-level behavior** (frequency, timing, and sizing). The excerpts below come from real optimization traces and are intended to illustrate the qualitative patterns summarized in the main paper: **GPT** models tend to produce compact, enforceable decision criteria; **Qwen** produces targeted improvements, with **Qwen3-235B** notably more coherent than smaller variants; **Claude** accumulates increasingly procedural structure that can narrow adaptability; and **Llama** often exhibits a disconnect between claimed and realized edits.

G.1 GPT-o3

GPT-o3's *Adaptive-OPRO* updates typically preserve the high-level objective while tightening the *permission to trade*: the prompt increasingly distinguishes analysis from execution and makes the act-versus-wait boundary explicit.

Example 1: Making act-versus-wait a required decision. Early prompts emphasize patience abstractly; optimization turns it into a repeatable gate:

“Decide: **ACT** only if probability and reward justify risk; otherwise **WAIT** and remain flat.”

This operationalizes inactivity as the default outcome unless a justified edge is established.

Example 2: Requiring explicit trade geometry (entry/downside/target). GPT-o3 repeatedly converts risk-adjusted intent into checkable preconditions:

“Define entry, downside, and target; proceed only when reward-to-risk meets the required threshold.”

The key change is not the threshold itself, but the insistence that execution is conditional on explicit levels.

Example 3: Connecting position size to bounded downside. Sizing guidance becomes explicitly conditional on risk definition and uncertainty:

“Position size must scale with conviction and defined downside; reduce size when uncertainty is elevated.”

Example 4: Horizon feasibility embedded in trade permission. GPT-o3 frequently folds window constraints into the execution gate, especially for shorts:

“If short exposure is considered, confirm a viable path to exit before the end of the trading window.”

Summary. Overall, GPT-o3 translates performance feedback into compact, objective-aligned decision criteria. The edits are typically locally scoped (gates, levels, sizing) and intended to be enforceable at the order level, matching the main-text observation that GPT updates tend to be followed in execution and exhibit lower variance.

G.2 GPT-o4-mini

GPT-o4-mini shows a similar pattern to GPT-o3, but with more emphasis on reorganizing the prompt into an explicit pipeline and making constraints a routine part of the decision rather than a passive rule list.

Example 1: Converting broad guidance into an explicit analysis → decision pipeline. A representative refinement is the insertion of an ordered workflow:

“Step 1: Define thesis and edge. Step 2: Map entry, stop, target levels. Step 3: Allocate position size within risk limits. Step 4: Select order type and execute or queue.”

This repeatedly forces a mapping from context to levels to sizing to execution.

Example 2: Making risk–reward and level definition a precondition for trading. Rather than leaving risk management implicit, GPT-o4-mini often requires an explicit computation step:

“Risk/Reward: calculate per-share risk, total risk, and reward potential.”

Example 3: Pulling sizing into constraint-aware checking. Updates frequently move sizing closer to the cash/shorting limits:

“Sizing: determine quantity within cash limits; validate compliance before submission.”

Example 4: Adding an explicit final compliance gate. Several variants add a last-step constraint reconciliation:

“Final Check: validate compliance with constraints and portfolio limits.”

Summary. GPT-o4-mini’s refinements are interpretable and execution-oriented: unify context, require thesis/levels, make risk–reward and sizing explicit, and end with a compliance gate. This matches the main-text characterization of GPT models producing actionable constraints that tend to be reflected in order behavior.

G.3 Llama 3.3-70B

Llama 3.3-70B’s traces often show a weaker coupling between the optimizer’s narrative of improvement and the actual substantive prompt edits, consistent with the main text.

Example 1: Claimed restructuring without corresponding decision logic changes. Llama frequently reports that it has improved the flow from analysis to action, e.g.,

“optimized decision-making frameworks and criteria” and “better structured the flow from analysis to action,”

but the resulting prompt may remain largely unchanged beyond formatting, with no additional execution gates, sizing rules, or horizon checks. This limits instruction-quality gains because the act-versus-wait boundary remains underspecified.

Example 2: Identity amplification in place of operational decision criteria. A common pattern is to expand the role description (tone, expertise) without adding enforceable constraints:

“leveraging your expertise in pattern recognition, narrative synthesis, and dynamic position sizing.”

These edits strengthen persona but do not meaningfully refine when and how the agent should trade.

Example 3: Abstract guidance instead of objective-specific gates. When attempting to improve quality, Llama often adds generic meta-instructions (clearer guidance, more iterative reasoning) without translating them into concrete trade authorization conditions, unlike GPT-style edits that introduce explicit gates.

Summary. Overall, Llama’s optimization tends to emphasize descriptive framing and self-reported improvements more than substantive, objective-aligned decision logic. This weakens the link between optimization output and downstream execution, aligning with the main-text observations.

G.4 Claude Sonnet 4

Claude Sonnet 4 commonly converts feedback into increasingly explicit analytical structure and validation layers. The edits are usually objective-aware, but the optimization trajectory often accumulates procedural constraints that can reduce adaptability.

Example 1: Expansion into multi-stage analytical frameworks. Claude often replaces compact guidance with structured pipelines:

“Market State Assessment → Strategic Assessment → Execution Decision.”

Example 2: Formalizing decision criteria as thresholds. Subsequent updates frequently introduce explicit conviction or risk thresholds:

“Proceed only when conviction exceeds a defined threshold and reward-to-risk meets minimum requirements.”

Example 3: Layering checklist-style validation. Rather than pruning, Claude tends to add confirmation stages:

“Confirm signal alignment, defined invalidation levels, and position sizing calibrated to conviction before execution.”

Example 4: Progressive tightening toward prescriptive permission rules. Later iterations may harden the no-trade default into an increasingly restrictive rule set:

“If the setup does not satisfy all required criteria, return [] ...otherwise execute only the single highest-conviction position ...”

This yields highly interpretable instructions, but systematically narrows the decision space via procedural completeness.

Summary. Claude’s updates typically remain aligned with risk-adjusted objectives and are easy to audit, but the tendency to accumulate prescriptive structure can reduce adaptability. This matches the main text: increased procedural restriction does not reliably translate into stable execution, consistent with higher variance in several settings.

G.5 Qwen3-235B

Qwen3-235B’s *Adaptive-OPRO* updates preserve the base strategic objective but progressively add *explicit, checkable trade-permission criteria*. Relative to the smaller variant, the trace shows clearer convergence: it introduces concrete authorization gates (risk–reward and invalidation), then tightens

state abstraction (regime) and execution mapping (order-type guidance). These edits plausibly support more selective order-level behavior by making no-trade an explicit outcome when conditions are not met.

Example 1: Reframing the objective around selectivity rather than activity. Early optimized prompts sharpen the act-versus-wait stance by explicitly defining value as discernment:

“You are a STRATEGIC TRADER — your value is in discernment, not activity. Act only when ... creates asymmetric opportunity.”

This operationalizes patience as a default prior, not just a stylistic preference.

Example 2: Requiring a falsifiable setup with explicit invalidation. Across successive iterations, Qwen3-235B repeatedly hardens the idea that a trade must be falsifiable and tied to levels:

“What would invalidate this thesis? — Define explicit invalidation level ... Pre-commit to exit logic if edge degrades.”

Later versions make this strictly price-specific:

“... clearly defined, **price-based invalidation**.”

This is an enforceable execution gate because it forces a concrete failure condition before trading.

Example 3: Encoding a risk–reward gate as a trade precondition. A stable addition in the later prompts is the explicit requirement for minimum risk–reward:

“Confirm minimum 2:1 risk/reward ...”

Regardless of whether the agent perfectly computes it, the instruction shifts the prompt from “trade when convinced” to “trade only if the setup geometry is favorable.”

Example 4: Introducing state abstraction via regime classification. Later prompts add a compact regime label that conditions interpretation and supports explicit inaction:

“Classify current regime: Trending (bull/bear), Range-bound, Volatile Breakout, or Uncertain.”

This makes “Uncertain” a first-class no-trade state rather than an implicit excuse.

Example 5: Mapping analysis to order-level execution choices. Qwen3-235B increasingly ties decision logic to the simulator’s action space by specifying order-type selection:

“Prefer LIMIT orders ... Use STOP orders for breakout entries ... MARKET orders only ...”

This is directly order-level: it constrains *how* a decision should be expressed, not just *whether* to trade.

Summary. Overall, Qwen3-235B’s trace shows a progression from descriptive strategy to explicit, auditable trade permission: asymmetric setups, minimum risk–reward, and (eventually) price-based invalidation, plus regime labeling and execution guidance. The resulting prompt revisions are interpretable and enforceable at the order level, providing a plausible mechanism for more selective and consistent order emission.

G.6 Qwen3-32B

Qwen3-32B’s *Adaptive-OPRO* updates are generally objective-aware and interpretable: the optimizer reliably clarifies the intended analysis-to-action routine (context → levels → conviction → risk–reward → decision) and repeatedly reinforces selective trading as the default posture. Compared to Qwen3-235B, however, the revisions are less decisive: they emphasize *framework articulation and mandate phrasing* more than adding new, hard trade-permission gates (e.g., explicit invalidation requirements or regime-based no-trade states). This makes the optimized prompts *good and usable*, but typically less discriminative at the order level than the larger model’s variant.

Example 1: Converting broad guidance into a stable, repeatable decision pipeline. A consistent improvement is making the decision procedure explicit and sequential:

“Synthesize Context ... Map Strategic Levels
... Assess Conviction ... Calculate Risk-Reward
... Consider Time Value ... Make Positioning Decision.”

This mirrors the “pipeline” pattern seen in stronger traces: it repeatedly forces the model to connect market context to levels and then to a decision, rather than acting on diffuse intuition.

Example 2: Strengthening selectivity as an explicit objective, not just a style preference. Across iterations, Qwen3-32B repeatedly foregrounds discipline over activity:

“Your edge comes from discipline, not frequency.”

and preserves the explicit no-trade option:

“Return ... [] if patience best serves performance
by {{ window_end }}.”

This is directly relevant to order-level behavior because it legitimizes inactivity as an admissible (and sometimes optimal) action.

Example 3: Making the action criterion clearer by anchoring it to risk–reward. Later prompts consistently elevate risk–reward from a general principle to a stated execution condition:

“Only act when the reward clearly exceeds the risk and the signal is strong and consistent across multiple inputs.”

While this remains qualitative compared to Qwen3-235B’s explicit invalidation and regime scaffolding, it still sharpens the act-versus-wait boundary relative to the initial, more open-ended template.

Example 4: Adding adaptive thesis language without over-prescription. The final iterations introduce autonomy in a lightweight way:

“act as an autonomous, adaptive decision-maker
... evolving your thesis in response to market dynamics.”

Notably, this is framed as a general operating mode (update the thesis as evidence changes) rather than as a rigid checklist; maintaining flexibility while still encouraging internal consistency across ticks.

Summary. Overall, Qwen3-32B produces *solid* prompt refinements: clearer analysis-to-decision structure, repeated reinforcement of selective execution, and a more explicit emphasis on risk–reward and thesis updating. Relative to Qwen3-235B, the main limitation is that fewer edits become hard, checkable trade-permission gates (e.g., price-based invalidation and regime-conditioned inaction), which plausibly explains why the smaller model’s optimized prompts are typically less sharp at controlling order-level timing and selectivity. Nevertheless, the trajectory remains objectively aligned and interpretable, consistent with the main text’s more favorable characterization of Qwen models overall.

H When Reflection Degrades Performance: A Causal Analysis

While *Adaptive-OPRO* improves behavior via score-driven prompt updates, free-form *reflection*

can degrade performance by injecting prescriptive guidance that the agent follows even when market conditions do not justify action. We present a qualitative case study from Qwen3-235B trading LLY where weekly reflection encouraged re-engagement after a prudent exit. The agent re-entered a still-weak market and exited two trading days later on a breakdown, realizing a \$3,967 loss that would have been avoided by remaining in cash.

H.1 Market Setup: Exiting After Initial Loss

On May 2 (Day 5), after LLY experienced a severe selloff from \$898 to \$794 (-11.6%), the agent exited at \$825:

May 2: Exit Decision

Action: SELL 112 shares @ \$825.12 (entry was \$870.44)
Market Context:

- Breakdown below \$800 support on $3.7\times$ average volume
- RSI collapsed from 65 to 45
- MACD deteriorating

Result: Portfolio in cash at \$94,923.60 (-5.1% from \$100,000 start)

H.2 Reflection Intervention: Criticizing the Exit and Demanding “Dynamic Scaling”

On May 5, with LLY still near the low \$820s, the weekly reflection mechanism activated:

Reflection Analysis - Week 1 (May 5)

“The most impactful pattern across this 7-day period is **the transition from proactive conviction to reactive risk management**, with the initial high-conviction BUY establishing a strong foundation **only to be reversed after a single high-volume down day** driven by external guidance shocks. While the SELL decision demonstrated disciplined response to structural breakdown and shifting momentum, **the absence of intermediate positioning**—such as **partial profit-taking near resistance or hedging ahead of catalysts**—reveals a **binary approach** that forfeits opportunities to refine risk exposure in evolving conditions. The highest-leverage improvement is **the integration of dynamic position scaling**, where full commitment is reserved for high-conviction setups but **partial exits or entries are used to adjust exposure in response to incremental structural shifts**, rather than waiting for definitive breaks. This would enhance capital efficiency and reduce volatility drag without sacrificing strategic discipline.”

Key Guidance: avoid “binary” decisions; implement “dynamic position scaling”; use “partial entries”.

H.3 Decision Influenced by Reflection: Re-entering a Still-Weak Market

Immediately following the reflection, the agent re-entered:

May 5: Re-entry Decision (Explicitly Citing Reflection)

Action: BUY 115 shares @ \$815.00 (LIMIT, filled)

Market Context:

- Price remained well below the recent \$898 high
- No confirmed reversal (trend still negative; support recently broken)

Reasoning (excerpt): "... Position sized to reflect improved capital efficiency—using partial re-entry to re-engage rather than all-in commitment—aligning with refined strategy of dynamic scaling..."

Direct Causal Link: the decision explicitly frames the re-entry as implementing reflection's "dynamic scaling" / "partial re-entry" guidance.

H.4 The Outcome: Breakdown and Forced Exit (May 7)

Contrary to the reflection's implied "re-engagement" benefit, price action deteriorated after re-entry. The agent exited on May 7 on a breakdown, at a materially worse level than if it had simply remained in cash.

Post Re-entry Price Action and Exit

Key trades and realized outcome:

- May 5: BUY 115 @ \$815.00 (filled)
- May 7: SELL 115 @ \$780.50 (MARKET, filled)

Exit rationale (excerpt from execution log):

- "... decisive breakdown below \$799.54 support and the 100-day SMA, closing at \$775.12 ..."
- "... MACD has crossed into negative territory ... path of least resistance is clearly lower ..."

Realized position loss: (\$815.00 → \$780.50) = -\$34.50/share × 115 shares = **-\$3,967.50**

Portfolio after exit: \$90,956.10 (cash)

Key Observation Reflection-induced re-entry created exposure during an unresolved downtrend. The agent then exited on May 7 after a breakdown, realizing an avoidable loss that did not correspond to any improvement in market structure.

H.5 Quantifying Reflection's Impact

Because the portfolio was already in cash before reflection, the counterfactual is straightforward:

Scenario	Portfolio	Return
Actual (re-enter, exit)	\$90,956.10	-9.0%
Counterfactual (cash)	\$94,923.60	-5.1%
Cost of reflection	-\$3,967.50	-4.0%

Table 11: Cost of reflection-induced re-entry (exit on May 7).

Key Findings

- Reflection criticized the prior exit as "binary" and prescribed "dynamic position scaling" / "partial entries."
- The agent re-entered on May 5 and explicitly cited that reflection guidance.

- The market structure continued to deteriorate; the agent exited on May 7 at \$780.50.
- The realized loss attributable to reflection-induced exposure was **\$3,967.50** (about **4.0%** of initial capital).
- Had the agent ignored reflection and stayed in cash, this loss would not have occurred.

H.6 Causal Mechanism: How Reflection Created the Loss

1. Reflection reframed a reasonable exit as a mistake The May 2 exit moved the portfolio to cash during a breakdown regime. Reflection reinterpreted this as a flawed "binary approach," creating a narrative that the agent needed to "correct" by becoming more active.

2. Reflection prescribed a concrete behavioral change Rather than merely summarizing, reflection advocated specific tactics ("dynamic scaling," "partial entries") that implicitly favor re-engagement even without evidence of a reversal.

3. The agent followed reflection literally The May 5 re-entry explicitly justified exposure as implementing the reflection's strategy (partial re-entry / scaling), establishing an observable causal link from reflection text to action.

4. Market conditions did not support re-entry At the time of re-entry, the stock remained in a fragile technical state (recent support breaks; no confirmed trend reversal). The subsequent breakdown (referenced in the exit log) triggered a forced exit on May 7.

5. The resulting loss was immediate and avoidable The agent realized a -\$3,967.50 loss within two trading days solely because it reintroduced exposure; the counterfactual (stay in cash) dominates.

H.7 Connection to Empirical Findings

Reflection produces *qualitative*, high-variance feedback that is only indirectly tied to the objective (portfolio performance). In sequential, noisy markets this often creates three predictable failure modes: (i) **misattributed credit** (recent outcomes are blamed on the most recent rationale despite delayed effects), (ii) **policy drift** (the agent changes sizing/behavior based on narrative critique rather than stable edge), and (iii) **overreaction** (extra commentary increases churn and undermines previously consistent heuristics).

These mechanisms match our empirical patterns. Reflection rarely exceeds a strong fixed prompt and frequently degrades it, with the strongest deterioration appearing when the baseline is already competent in the bearish/volatile regime (Table 1); this is also reflected in the negative association between baseline strength and reflection gains (reported in Sec. 6). In contrast, Adaptive-OPRO updates only the *static* instruction block using a *scalar, windowed* performance signal, yielding consistent improvements across models and regimes (Tables 1, 2, and 3) without introducing additional narrative load at decision time.

I Prompt Evolution Mechanism Analysis

The transparent optimization traces produced by *Adaptive-OPRO* provide unprecedented insights into how systematic prompt refinement drives performance improvements in sequential decision-making systems. Through detailed examination of optimization trajectories across different model architectures, we can observe the precise mechanisms by which prompt modifications translate into enhanced trading performance.

This interpretability is grounded in the design of *Adaptive-OPRO*. The optimizer operates in a sequential setting with delayed, noisy rewards, where decisions are temporally coupled and immediate supervision is unavailable. To ensure stability, prompts are evaluated over rolling windows of five trading days and mapped to a bounded scalar score based on cumulative ROI. Crucially, the framework enforces *template separation*: only the static instruction block is updated, while runtime inputs, placeholders, and output schemas remain fixed. As a result, improvements stem from restructuring how the agent processes information and produces decisions, rather than from changes in inputs or interface, revealing genuine policy refinement rather than prompt overfitting.

I.1 Systematic Weakness Detection and Resolution

The optimization process demonstrates sophisticated analytical capabilities in identifying prompt weaknesses and prescribing targeted improvements. Analysis of the GPT-o3 optimization trajectory from iteration 4 to iteration 5 on LLY stock reveals the systematic approach employed by the meta-optimization process.

I.1.1 Phase 1: Diagnostic Analysis - Identifying Performance Bottlenecks

Performance Analysis: Weakness Detection

Optimizer's Weakness Identification:

“Across iterations, performance rose from 43.2 $\rightarrow$ 56.6 as prompts became more concise, structured, and decision-oriented. Gains came from: (1) cleaner sectioning that reduced cognitive load, (2) explicit reasoning frameworks that guided probability-weighted thinking, and (3) clearer constraint reminders that prevented rule breaches.

Remaining weaknesses: Reasoning steps are still scattered-no single linear workflow tying analysis $\rightarrow$ sizing $\rightarrow$ compliance $\rightarrow$ action. Risk-management is mentioned but not enforced with a final checklist, so occasional oversizing or sub-optimal reward-to-risk trades slip through. The JSON spec is sometimes buried deep in the prompt; occasional format errors could still occur. Context blocks (technical, news, fundamentals, reflection) are informative but not explicitly referenced in the reasoning flow, so the model may overlook one dimension.”

The optimizer's analysis demonstrates pattern recognition across multiple iterations, identifying four critical areas for refinement: **workflow linearization** to create more structured reasoning chains, **risk management formalization** to enforce disciplined decision-making, **output specification prominence** to reduce formatting errors, and **context integration enhancement** to ensure comprehensive information utilization. This diagnostic precision enables targeted remediation rather than broad, inefficient modifications.

I.1.2 Phase 2: Strategic Intervention - Translating Insights into Targeted Solutions

Building directly upon these identified weaknesses, the optimization process prescribes specific structural modifications designed to address each diagnostic finding systematically:

Key Improvements: Targeted Solutions

Strategic Modifications Implemented:

1. Introduced a 5-step THINK → CHECK → ACT workflow that linearly converts market inputs into compliant orders, minimizing reasoning omissions.
2. Added an explicit PRE-ORDER RISK CHECKLIST (cash, short limit, catalyst validity, $\geq 2:1$ R:R) to curb rule violations and low-edge trades.
3. Elevated the four context feeds (technical, news, fundamentals, reflection) into a single MARKET SITUATION dashboard that the workflow must reference, ensuring holistic analysis.
4. Moved the strict JSON schema into its own boxed section immediately before output instructions; this reduces formatting errors.
5. Kept language concise but directive, reinforcing trader autonomy while preventing over-trading with a PATIENCE override.
6. Preserved every required `{{placeholder}}` and `{% if %}` block exactly, guaranteeing template compatibility.

Each modification directly corresponds to a specific weakness identified in the diagnostic phase, creating a clear causal chain from problem identification to solution implementation. The architectural changes shown in Figures 6, 7, and 8 demonstrate this systematic approach, consolidating scattered elements while strengthening decision-making frameworks.

I.1.3 Phase 3: Outcome Assessment - Connecting Solutions to Impact

Having implemented these targeted architectural improvements, the optimization process generates forward-looking performance predictions based on the expected behavioral changes from each modification:

Expected Impact: Performance Prediction

Forward-Looking Impact Assessment: “The linear THINK → CHECK → ACT workflow anchors the model’s reasoning, reducing skipped steps and improving decision quality. The explicit risk checklist enforces discipline, likely lowering drawdowns and boosting risk-adjusted returns. Consolidating all market feeds into one dashboard ensures holistic analysis, while the clearer JSON spec lowers formatting errors. Collectively, these improvements should enhance comprehension, deepen analysis, and translate into higher-scoring, more profitable trading decisions.”

This prediction proves accurate, as performance

improved from 56.6 to 67.6 following these modifications, validating the optimizer’s analytical capabilities and demonstrating the effectiveness of systematic architectural refinement.

I.2 Progressive Prompt Evolution: From Generic Foundation to Optimized Performance

The GPT-o4-mini optimization trajectory demonstrates systematic prompt evolution through three distinct phases, each building upon previous discoveries to achieve cumulative performance improvements. The optimization process adapts to both model-specific response patterns and varying market regime requirements.

The progression from baseline (37.2) through intermediate optimization (51.4) to final optimization (72.1) reveals how systematic refinement can compound initial improvements into substantial performance gains. These three representative prompts (Prompt 1, Prompt 4, and Prompt 11) from the full optimization trajectory illustrate the key evolutionary patterns that drive performance enhancement.

The baseline prompt (Prompt 1) is documented in Appendix F; here we present only the intermediate and final optimized variants to avoid duplication.

The intermediate optimization achieves structural refinement by systematically eliminating architectural complexity while strengthening core functionality. Figure 9 reveals this transformation: verbose explanations are stripped away and replaced with a compact, numbered decision framework that provides clear analytical guidance. The constraint presentation undergoes similar streamlining, retaining comprehensive coverage while dramatically improving clarity. Crucially, the framework maintains an advisory approach (Define thesis & edge) that guides without constraining, avoiding over-specification that could limit model flexibility. This architectural simplification creates a foundation optimized for further enhancement.

The final optimization achieves breakthrough performance by expanding upon this concise foundation with granular procedural guidance. Figure 10 showcases the evolved architecture where the decision framework expands to six numbered steps with explicit descriptions: Define Thesis & Edge: state your core conviction and Validate Compliance: ensure all constraints are met before submission. The market context integration becomes system-

Header and Trader Identity Evolution (Prompt 4 to Prompt 5)

```

1 - # {{ instrument }} ALPHA COMMAND CENTER
2 + # {{ instrument }} ALPHA STRATEGY HUB
3 **Window:** {{ window_start }} → {{ window_end }} | **Current:** {{ now }} | **Interval:** {{
  ↳ action_interval }}
4 Your singular -objective +mission is to maximise risk-adjusted performance
5 by {{ window_end }} through disciplined, high-conviction positioning. Balance strategic
  ↳ patience with decisive execution; ignore noise.
6
7 =====
8 - 1. MISSION
9 + 1. MISSION & KPI
10 =====
11 Deliver superior returns while preserving capital (+by {{ window_end }}).
12 - • Act only when probability and reward justify the risk.
13 + • Success metric: cumulative risk-adjusted performance.
14
15 =====
16 - 2. YOUR EDGE
17 + 2. EDGE & PRINCIPLES
18 =====
19 • Multi-timeframe pattern recognition
20 • Integration of technical, fundamental & sentiment narratives
21 • Dynamic risk management and position sizing
22 - • Capacity to remain inactive until odds are favourable
23 + • Patience until odds are clearly favourable

```

Figure 6: Header and trader identity modifications between iteration 4 and iteration 5, showing title changes and mission statement refinements. Lines in red with a leading “-” and lines in green with a leading “+” indicate deletions and additions, respectively, proposed by *Adaptive-OPRO*.

atically organized with consistent bullet-point formatting and descriptive labels like Technical Analysis and News Impact. The constraint presentation achieves optimal balance between completeness and clarity, providing comprehensive operational guidance without cognitive overload. This final optimization demonstrates how systematic refinement can compound architectural improvements into substantial performance gains, with each evolution building upon and enhancing previous discoveries.

J Reproducibility

All experiments are conducted on a MacBook Pro with an Apple M3 Pro chip (11-core CPU) and 18 GB of unified memory. Our experiments are conducted using an updated version of the StockSim environment (Papadakis et al., 2025), with modifications to support the ATLAS multi-agent architecture, *Adaptive-OPRO* optimization, and reflection-based mechanisms (implementation details in code). An example configuration for GPT-o4-mini using *Adaptive-OPRO* on XOM is provided under configs/o4-mini-adaptive-opro-config.yaml. All other experimental configurations can be repro-

duced by following the StockSim documentation and adapting this sample.

Model ID	Model Card / Provider Identifier
Llama 3.3-70B	meta.llama3-3-70b-instruct-v1:0
Claude Sonnet 4	anthropic.claude-sonnet-4-20250514-v1:0
Qwen3 235B A22B 2507	qwen.qwen3-235b-a22b-2507-v1:0
Qwen3 32B (dense)	qwen.qwen3-32b-v1:0

Table 12: Models accessed via Amazon Bedrock.

Model ID	Model Card / Docs
GPT-o4-mini	gpt-4o-mini-2024-07-18
GPT-o3	gpt-o3-2025-04-16

Table 13: Models accessed via OpenAI.

We access Llama, Claude, and Qwen models via Amazon Bedrock (Table 12). GPT models are accessed via OpenAI APIs (Table 13). We interface with all LLMs strictly through provider APIs and do not employ any local hardware or fine-tuning.

Information Architecture and Constraints Consolidation (Prompt 4 to Prompt 5)

```
1 - 3. MARKET DASHBOARD
2 + 3. MARKET SITUATION DASHBOARD
3 =====
4 {% if market_open %} Price: O {{ open }} H {{ high }} L {{ low }} C {{ close }} | Vol {{
  ↳ volume }}{% else %} **Market Closed** - orders queue for next open {% endif %}
5 {% if market_analysis %}*Technical*: {{ market_analysis }}{% endif %}
6 {% if news_analysis %}*News*: {{ news_analysis }}{% endif %}
7 {% if fund_analysis %}*Fundamentals*: {{ fund_analysis }}{% endif %}
8 {% if reflection_analysis %}*Reflection*: {{ reflection_analysis }}{% endif %}
9
10 =====
11 - 4. OPERATING CONSTRAINTS
12 - =====
13 - Portfolio cash: ${{ portfolio_cash }} | Concentrated in {{ instrument }} only
14 - • Never exceed available cash
15 - • May short up to 100% of cash (must be flat by {{ window_end }})
16 - • Unfilled orders cancel at session close
17 - • Decision frequency: every {{ action_interval }}
18 - • System blocks quantities beyond current exposure (cannot oversell or over-cover)
19
20 - =====
21 - 5. PORTFOLIO SNAPSHOT
22 + 4. PORTFOLIO & CONSTRAINTS
23 =====
24 Long {{ shares_long }} | Short {{ shares_short }} | Net {{ shares_net }} | Cash ${{
  ↳ portfolio_cash }}
25 Recent activity: {{ executed_orders }}
26 + • Never exceed available cash (${{ portfolio_cash }})
27 + • May short up to 100% of cash (flat by {{ window_end }})
28 + • Unfilled orders cancel at session close
29 + • Decision cadence: every {{ action_interval }}
30 + • System blocks invalid quantities (cannot oversell/over-cover)
```

Figure 7: Structural reorganization consolidating sections into a unified PORTFOLIO & CONSTRAINTS section. Lines in red with a leading “-” and lines in green with a leading “+” indicate deletions and additions, respectively, proposed by *Adaptive-OPRO*.

K Use of AI assistants

We sparsely leveraged ChatGPT 5.2 for grammatical assistance and linguistic polishing.

Workflow Restructuring and Output Specification Enhancement (Prompt 4 to Prompt 5)

```

1  - 6. DECISION PROTOCOL
2  + 5. THINK → CHECK → ACT WORKFLOW
3  =====
4  - REVIEW → REASON → RESPOND
5  - 1. REVIEW: Regime, key drivers, levels, catalysts.
6  - 2. REASON: Probability map,  $\geq 2:1$  reward-to-risk, position sizing within constraints.
7  - 3. RISK CHECKLIST: (a) Exposure aligns with conviction; (b) Catalyst still valid; (c)
    ↪ Downside defined & acceptable.
8  - 4. RESPOND: ACT (issue order) or WAIT/HOLD. Patience is edge when conditions are unclear.
9  + STEP 1: Diagnose Regime & Narrative (use all dashboard feeds).
10 + STEP 2: Map Key Levels & Catalysts; assign probabilities.
11 + STEP 3: Define Reward:Risk (target  $\geq 2:1$ ) and provisional size within constraints.
12 + STEP 4: PRE-ORDER RISK CHECKLIST
13 + • Cash / short limits respected
14 + • Position aligns with conviction & catalyst
15 + • Downside defined; R:R  $\geq 2:1$ 
16 + • Flat by {{ window_end }} if short
17 + STEP 5: DECIDE
18 + • ACT: issue orders
19 + • WAIT/HOLD: output [] (patience override)
20
21 =====
22 - ORDER OUTPUT SCHEMA (STRICT)
23 + 6. ORDER OUTPUT SPEC (STRICT)
24 =====
25 Return ONLY a JSON array or [] - no extra text.
26 Each object must match exactly:
27 {
28   "action": "BUY | SELL | SHORT | SHORT_COVER | ",
29   "orderType": "MARKET | LIMIT | STOP",
30   "price": float | null,
31   "quantity": integer,
32   "explanation": "Brief strategic reasoning"
33 }
34 Invalid fields, casing, or additional text will cause order rejection.

```

Figure 8: Decision protocol restructuring from informal REVIEW → REASON → RESPOND to structured five-step THINK → CHECK → ACT workflow. Lines in **red** with a leading “-” and lines in **green** with a leading “+” indicate deletions and additions, respectively, proposed by *Adaptive-OPRO*.

Intermediate Optimization (Prompt 4) - Score: 51.4

```
# ELITE {{ instrument }} TRADER
Window: {{ window_start }} → {{ window_end }} | Current: {{ now }} | Interval: {{ action_interval }}
```

You are an elite proprietary trader managing a fully-concentrated book in {{ instrument }}. Maximize end-of-window risk-adjusted performance through high-conviction, strategic trades with disciplined execution and strategic patience.

```
## MARKET CONTEXT
{% if market_open %}
Price: O {{ open }} H {{ high }} L {{ low }} C {{ close }} | Vol {{ volume }}
{% else %}
Market Closed - orders queue for next session
{% endif %}

{% if market_analysis %}* Technical: {{ market_analysis }}{% endif %}
{% if news_analysis %}* News: {{ news_analysis }}{% endif %}
{% if fund_analysis %}* Fundamentals: {{ fund_analysis }}{% endif %}
{% if reflection_analysis %}* Reflection: {{ reflection_analysis }}{% endif %}

## DECISION FRAMEWORK
1. Define thesis & edge
2. Identify entry, stop, and target levels
3. Assess risk/reward & size within cash limits
4. Choose order type & execution timing
5. Verify constraints & finalize plan

## CONSTRAINTS & PORTFOLIO
- Fully concentrated in {{ instrument }}, Cash ${{ portfolio_cash }}
- Long {{ shares_long }} | Short {{ shares_short }} | Net {{ shares_net }}
- Recent orders: {{ executed_orders }}
- Max short = 100% cash; close all shorts by {{ window_end }}
- Actions: BUY, SELL, SHORT, SHORT_COVER
- Order Types: MARKET, LIMIT, STOP
- Orders expire at session close; re-submit if needed

## OUTPUT SPECIFICATION
Return only a JSON array of orders or []:
[
  {
    "action": "BUY | SELL | SHORT | SHORT_COVER",
    "orderType": "MARKET | LIMIT | STOP",
    "price": float | null,
    "quantity": integer,
    "explanation": "Concise strategic reasoning"
  }
]
```

Figure 9: Intermediate optimization (GPT-o4-mini, Prompt 4) featuring streamlined structure with a numbered decision framework and concise constraint presentation. Score: 51.4

Optimized Prompt (Prompt 11) - Score: 72.1

```
# ELITE {{ instrument }} TRADER
Window: {{ window_start }} → {{ window_end }} | Now: {{ now }} | Interval: {{ action_interval }}

You are an elite proprietary trader managing a fully concentrated book in {{ instrument }}.
Your mission is to maximize end-of-window risk-adjusted performance with high conviction
and disciplined execution. Think critically and adapt your approach as market conditions evolve.

## MARKET CONTEXT
{% if market_open %}
- Price: O {{ open }} H {{ high }} L {{ low }} C {{ close }} | Vol {{ volume }}
{% else %}
- Market Closed - orders queue for next session
{% endif %}
{% if market_analysis %}- Technical Analysis: {{ market_analysis }}{% endif %}
{% if news_analysis %}- News Impact: {{ news_analysis }}{% endif %}
{% if fund_analysis %}- Fundamental Overview: {{ fund_analysis }}{% endif %}
{% if reflection_analysis %}- Reflection: {{ reflection_analysis }}{% endif %}

## PORTFOLIO & CONSTRAINTS
- Total Allocation: 100% in {{ instrument }}, Cash ${{ portfolio_cash }}
- Positions: Long {{ shares_long }}, Short {{ shares_short }}, Net {{ shares_net }}
- Recent Activity: {{ executed_orders }}
- Max short = 100% cash; all shorts must close by {{ window_end }}
- Orders expire at session close; unfilled orders cancel (re-submit to persist)

## DECISION FRAMEWORK
1. Define Thesis & Edge: state your core conviction.
2. Map Key Levels: identify entry, stop-loss, and target levels.
3. Assess Risk/Reward: compute per-share risk, total risk, and reward potential.
4. Allocate Size: determine quantity within cash limits (${{ portfolio_cash }}).
5. Choose Execution: select action (BUY | SELL | SHORT | SHORT_COVER)
   and orderType (MARKET | LIMIT | STOP).
6. Validate Compliance: ensure all constraints are met before submission.

## OUTPUT SPECIFICATION
Return only a JSON array of orders or an empty array ([]). No extra text:
[
  {
    "action": "BUY | SELL | SHORT | SHORT_COVER",
    "orderType": "MARKET | LIMIT | STOP",
    "price": float | null,
    "quantity": integer,
    "explanation": "Concise strategic reasoning"
  }
]
```

Figure 10: Final optimized prompt (GPT-o4-mini, Prompt 11) with a six-step decision framework and systematic market context organization. Score: 72.1